The Labor Market Impacts of GLP-1s

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Abstract

We estimate the causal impacts of GLP-1 treatment on labor market outcomes using linked Danish administrative data and a matched stacked difference-in-differences design. We compare patients who initiate GLP-1 treatment during the first two years of Semaglutide availability to observably similar patients who initiate four years later. We find that GLP-1 treatment reduces long-term sickness leave days by 22.8%, leading to total economic benefits of approximately 1.5–2.2% of annual labor income per employed individual. We do not detect statistically significant or economically meaningful impacts on employment, labor force participation, or total income over four years.

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1 Introduction

Obesity and type 2 diabetes are two of the most prevalent chronic conditions in developed countries. Across OECD countries, almost one in four adults is obese, and diabetes affects 7% of adults (OECD, 2019, 2023). Both conditions are associated with symptoms that may reduce productivity, increase absenteeism and presenteeism, and weaken labor market attachment. Type 2 diabetes is associated with fatigue, hypoglycemia, neuropathy, and depression (Romadlon et al., 2022; Wang et al., 2019; Mori et al., 2019), while obesity is associated with pain, mobility limitations, and obstructive sleep apnea (Mamun et al., 2026; Zhang et al., 2018; Foster et al., 2009). GLP-1 receptor agonists have emerged as a promising treatment for both conditions,[1] with clinical evidence suggesting that they can alleviate several of these limitations soon after initiation (Wu et al., 2026; Bliddal et al., 2024; Abd El Aziz et al., 2017; Blackman et al., 2016). Whether these clinical improvements translate into better labor market outcomes remains unclear. In this paper, we study the causal effects of GLP-1 treatment on labor market outcomes.

Despite the rapid diffusion of GLP-1 drugs, rigorous causal evidence on their labor market impacts remains scarce for at least two reasons. The first is selection into treatment, as patients who initiate GLP-1 treatment differ from non-initiators in observed and potentially unobserved characteristics that may also relate to labor market outcomes. The second is lack of data. Addressing this question requires linked longitudinal records that follow the same individuals across prescription histories, health outcomes, and labor market trajectories over a sufficiently long horizon to detect downstream effects.

Our work overcomes these challenges using linked Danish administrative data and a matched stacked difference-in-differences design. We compare patients who initiate GLP-1 treatment during the first 6 quarters of availability (2018 Q3 to 2019 Q4) to observably similar patients who initiate treatment 4 years later (2022 Q3 to 2023 Q4).[2] To improve comparability, we first exact-match patients on key baseline characteristics, then implement nearest-neighbor propensity score matching within exact-match strata. This yields a matched sample of 7,021 early initiators and 7,021 later

[1] GLP-1 receptor agonists are a class of drugs that mimic endogenous glucagon-like peptide-1 to stimulate insulin secretion, suppress glucagon release, and at higher doses promote weight loss through appetite regulation.

[2] Within the GLP-1 class, Semaglutide represented a substantial advance over earlier agents in terms of glycemic control, weight loss, and cardiovascular risk reduction (Marso et al., 2016). Semaglutide is marketed under two brand names targeting different indications: Ozempic for type 2 diabetes and Wegovy for obesity. We focus on Ozempic as it provides the longest available follow-up for Semaglutide treatment in Denmark. Ozempic became available in August 2018, while Wegovy was not introduced until December 2022.

initiators. We then estimate a stacked difference-in-differences event study that traces outcomes over a four-year horizon.

We begin by documenting a strong and persistent first stage in GLP-1 use. Four years after initiation, 75% of treated patients remain on GLP-1 treatment. This level of persistence in treatment exposure increases the scope for detecting downstream effects over a four-year horizon. Next, we document reductions in emergency department visits, general practitioner visits, and cardiovascular medication use, consistent with an improvement in patient health.

We next examine whether GLP-1 treatment reduces long-term sickness leave, defined as spells exceeding 30 days. We find that treatment reduces sickness leave days by 0.864 days, corresponding to a 22.8% decline relative to the pre-initiation mean of 3.789 days per quarter. The effect grows from 0.809 days in the first two post-initiation years to 0.912 days in years three and four. These estimated impacts on long-term sickness leave imply annual combined employer, public, and individual benefits of about DKK 4,437 (approximately 1.5% of annual labor earnings or USD 672) per treated employed individual, and up to DKK 6,557 (2.2%) if short-term absences decline proportionally. This highlights reduced sickness leave as an economically meaningful channel through which these drugs generate value beyond their direct health effects.

Despite finding a meaningful decline in long-term sickness leave, we estimate statistically precise null effects on employment, labor force participation, and total income over a four-year horizon. We can rule out increases in employment of more than 0.5 pp and increases in total income of more than 1.6%. These null results also extend across heterogeneity margins, as no subgroup shows a statistically significant gain in employment, labor force participation, or total income. Because the Danish sickness benefit system provides near-full income replacement for much of our sample, reductions in sickness absence primarily generate employer and public savings rather than higher measured income for workers. In settings with less complete sickness insurance, such as the United States, similar health improvements may appear more directly in earnings. Our design cannot separate whether the null effects reflect this institutional setting, health gains too small to shift employment, or the composition of early initiators.

Our paper contributes to the literature on the economic effects of medical treatments. Prior work has documented labor market benefits of treatments for chronic pain (Garthwaite, 2012; Bütikofer and Skira, 2018), mental health conditions (Shapiro, 2022; Biasi et al., 2025; Bhalo-

tra et al., 2025), and cancer (Daysal et al., 2025).[3] We add to this literature by documenting the labor market effects of a class of drugs that targets obesity and diabetes, two conditions that affect more than 800 million adults worldwide. We also contribute to the emerging economics literature on GLP-1 drugs. Existing work has focused primarily on medical spending and clinical outcomes. Bock et al. (2026) and Wing et al. (2026) study the effects of GLP-1 treatment on healthcare spending and find no statistically significant reduction in spending over a four- to five-year horizon, although Bock et al. (2026) document a decline in major adverse cardiovascular events among patients with pre-existing cardiovascular disease. Recent work has begun to examine broader non-health outcomes of GLP-1 treatment, with mixed findings across settings and outcomes. Kaestner and Schiman (2026) study adults with diabetes in the Medical Expenditure Panel Survey and find little evidence of changes in mental health, self-rated health, employment, or marriage over short horizons of one to two years. Diamond (2026) studies GLP-1 use for weight loss among women in the Understanding America Study and finds large increases in employment among women who were not employed at baseline, and in marriage or cohabitation among women who were single at baseline. We add to this emerging literature by studying GLP-1 treatment in population-wide Danish administrative data that link prescription histories to health care utilization, medically certified sickness leave, and labor market outcomes. We document high adherence to GLP-1 treatment over four years and show that sustained exposure reduces medically certified sickness leave and generates meaningful fiscal spillovers.

2 Institutional Setting

2.1 GLP-1 Receptor Agonists in Denmark

Denmark has a tax-financed universal health insurance system in which general practitioners (GPs) serve as the primary point of contact for chronic disease management, including type 2 diabetes. Most patients with type 2 diabetes receive treatment in general practice, with referrals to endocrinologists or diabetologists reserved for complex or treatment-resistant cases (Mailhac et al., 2024). Treatment of type 2 diabetes ranges from lifestyle modification to pharmacotherapy, with common drug classes including biguanides (metformin), SGLT2 inhibitors, DPP-4 inhibitors, sulfonylureas,

[3] A related literature studies how health and wellness interventions affect productivity and labor market outcomes more broadly (Stephens and Toohey, 2022; Jones et al., 2019).

and insulin. Ozempic became available in Denmark in August 2018.[4] Throughout our study period, Ozempic was indicated for adults with type 2 diabetes whose condition was inadequately controlled through diet, exercise, and prior antidiabetic medication. It was prescribed either in place of a metformin prescription if deemed unsuitable, or as an add-on therapy to existing diabetes treatment (Medicin.dk, 2026a). Eligible prescriptions are subsidized: adult patients face a sliding copayment schedule with a maximum annual out-of-pocket amount of DKK 4,300 in 2019 (approximately USD 645), while a one-month supply of Ozempic in 2019 was DKK 1,117 (approximately USD 167) (Medicinpriser, 2026).[5]

2.2 Long-Term Sickness Leave in Denmark

In Denmark, long-term sickness leave is covered through the sickness benefit system. Eligibility generally requires that the individual be unable to work because of their illness and that they have sufficient labor market attachment. Employees, self-employed individuals, and unemployed individuals who would otherwise be entitled to unemployment insurance benefits may qualify under separate eligibility rules. Salaried employees keep their usual salary as long as the illness and the employment last. Hourly workers covered by a collective agreement typically receive full pay for 9–14 weeks followed by the statutory benefit, while those outside an agreement receive only the statutory benefit.[6] Employers fully cover the first 30 calendar days, after which the municipality pays the statutory benefit, either as partial reimbursement to the employer or directly to the worker. The self-employed receive the statutory benefit after a 14-day waiting period, while the unemployed receive the same amount as their unemployment benefits, paid by their fund for the first 14 days and by the municipality thereafter. Regardless of labor market status, the statutory benefit is paid for at most 22 weeks within 9 months, the shortest duration in the OECD, where the average benefit duration is one year (Lee et al., 2025).[7] Employees on sickness leave can be dismissed with regular notice, or with a substantially reduced notice if paid sick leave reaches 120 days within twelve

[4] Appendix Figure B.1 shows the number of patients using any Semaglutide, Ozempic, Wegovy, or Rybelsus over time. Wegovy was introduced in Denmark in December 2022.

[5] This is for a four-week pack of 0.25 mg. Prices declined substantially over the study period: from DKK 1,339 in 2018 to DKK 1,117 in 2019 and DKK 213 by 2026.

[6] The statutory benefit is earnings-related but capped at DKK 4,355 per week (approximately USD 2,837 per month) in 2019 (Beskæftigelsesministeriet, 2026). For an average-wage worker, it replaces 54% of earnings, compared with an OECD average of 63% (OECD, 2020). However, most employees receive full pay, as around 82% of them are covered by a collective bargaining agreement (OECD and AIAS, 2025).

[7] Sickness benefits can be extended under specific conditions, including severe illness, ongoing medical treatment, and the need for further assessment of work capacity.

months. Overall, Denmark ranks in the lowest third of the OECD dismissal-protection indicator, well below the OECD median but above the UK and the US (OECD, 2026).

3 Data and Variables

We use several population-wide Danish administrative registers that we link at the individual level using unique personal identifiers. As detailed in Section 4, we compare early Ozempic initiators to matched later initiators in a stacked difference-in-differences design. This section describes measurement of treatment initiation, treatment persistence, outcomes, and baseline covariates. Further details are provided in Appendix A.

GLP-1 prescriptions. We obtain Semaglutide prescriptions from administrative records covering the universe of filled prescriptions in Denmark. We define treatment initiation as the date of an individual's first Ozempic prescription, requiring that this also be the individual's first Semaglutide prescription. We then construct an indicator for any Semaglutide prescription fill in each quarter. This measure serves as our first stage outcome and captures persistence in treatment after initiation.

Primary Outcomes. We measure the number of sickness leave days in a quarter in spells lasting more than 30 days. We study other measures of sickness leave in the appendix, including share of months with sickness leave in a quarter, an indicator for any long-term sickness leave in a quarter, and municipality-paid sickness leave benefits (after the first 30 days[8]) from annual tax records, and short-term sickness absences for employees in the public sector and in large private firms.[9]

Our labor market outcomes capture yearly employment, labor force participation, and total income. We define employment and being out of the labor force from annual data on labor force status at the end of November. We obtain annual total income from tax records. Total income includes earned income, income from self-employment, sickness benefits, other public transfers, and other/residual income; we also study these individual components in the appendix. All monetary values are expressed in 2025 DKK.

[8] For individuals covered by unemployment insurance, eligibility begins after 14 days.

[9] In a complementary data set, we observe all sickness absences, including those lasting fewer than 30 days, for employees in the public sector and in private firms with more than 250 employees. Since only 39% of our sample is employed in such firms at baseline, we use these data for robustness checks and supplementary economic benefit calculations rather than for the main analysis.

Additional Outcomes. To provide supplementary context on healthcare use around GLP-1 treatment, we also construct three healthcare utilization outcomes in a given quarter.[10] The first is an indicator for any emergency department encounter, drawn from hospital records. The second is an indicator for any cardiovascular drug prescription, extracted from prescription records using ATC code C. The third is the number of GP services received, obtained from national health insurance claims data.

Baseline Covariates. We define type 2 diabetes using hospital contacts with an International Classification of Diseases (ICD-10) diagnosis code of E11 or filled prescriptions for diabetes medication. We define obesity using hospital contacts with an ICD-10 diagnosis code of E66, excluding E66.3 (overweight).[11] We obtain marital status and region of residence from population records, highest educational attainment from education records, occupation and industry codes from the labor market register, and the number of GP services and total hospital expenditures from national health insurance claims data. Diabetes and obesity status are measured in 2016–2017 while other baseline covariates are measured in December 2017.

4 Empirical Strategy

To estimate the causal effects of GLP-1 treatment on labor market outcomes, we implement a stacked difference-in-differences design on a matched sample of observably similar early and later initiators. We begin by constructing a treated group consisting of patients who first initiate Ozempic treatment during the first two calendar years of availability, from August 2018 through December 2019. The comparison group consists of patients who first initiate treatment four years later, from August 2022 through December 2023. This early-versus-later initiator design allows us to trace treatment effects over a four-year horizon. This comparison is similar in spirit to the designs used in Wing et al. (2026) and Diamond (2026) and is implemented using a stacked difference-in-differences estimator (Cengiz et al., 2019; Deshpande and Li, 2019; Conti et al., 2025).

A central challenge with an early vs later treated comparison is that GLP-1 adoption expanded rapidly over a short period (as shown in Appendix Figure B.1), so early and later initiators differ

[10] These measures capture healthcare utilization and medication management, which may reflect health behavior in addition to underlying health. Our data do not contain direct clinical measures, such as biomarkers.

[11] We only observe diagnoses associated with hospital contacts, therefore we expect underlying rates of diabetes and obesity to be higher than indicated in the data.

systematically in observed characteristics. Prior studies show that early initiators in Denmark are older and more likely to have diabetes than later adopters (Mailhac et al., 2024).[12] These differences raise concerns about comparability between early and later initiators. For example, early initiators may disproportionately consist of patients starting GLP-1 therapy to manage treatment-resistant diabetes and related comorbidities, whereas later initiators may be younger and healthier patients seeking treatment primarily for weight loss. If these groups would have followed different health and labor market trajectories even in the absence of treatment, a simple comparison of early and later initiators would yield biased estimates of GLP-1's causal effects.[13]

To improve comparability, we match early initiators to later initiators (who initiate 4 years later) using nearest-neighbor matching within exact-match strata (Ho et al., 2017; Jäger et al., Forthcoming; Schmieder et al., 2023; Bertheau et al., 2023).[14] We first exact-match patients on four key variables in 2018: sex, age (in five-year bins), diabetes, and obesity status—the latter two measured over 2016–2017. We match on fixed 2018 characteristics because they provide a common pre-treatment baseline, whereas pre-initiation characteristics may already reflect endogenous changes related to treatment timing. Within these strata, we then one-to-one match early and later initiators on the propensity score estimated from marital status, educational attainment, region of residence, income, number of general practitioner services, hospital expenditures, labor force status, industry sector, and occupation, all measured in November 2017.

This hybrid procedure balances close comparability on key predictors of GLP-1 initiation, health risk, and life-cycle labor market outcomes against preservation of common support and the match rate. We exact-match on sex, age, diabetes, and obesity, using five-year age bins to retain enough comparable later initiators for each early initiator. Within these strata, nearest-neighbor propensity score matching selects later initiators who are similar to early initiators on a richer set of observed baseline covariates, including economic and health measures. We match only on cross-sectional baseline characteristics, rather than on pre-treatment trends.

Equipped with the matched sample of early and later initiators, we estimate a stacked

[12] Columns 1 and 2 of Table 1 show the same pattern in our data. Bock et al. (2026) document a similar pattern in the U.S. Veterans Health Administration.

[13] As previously mentioned, we focus on initiators of Ozempic (as opposed to *e.g.*, Wegovy) to improve comparability of early vs later treated. Wegovy became available in Denmark in December 2022 to primarily manage weight loss; early initiators used Ozempic for diabetes and weight management.

[14] Zeltzer et al. (2023) employ a similar method, using Euclidean distance to determine the nearest-neighbor within an exact-match stratum of a telehealth device adopter (vs non-adopter); Altındağ et al. (2026) employ propensity score within exact-match strata, with replacement.

difference-in-differences event study. Specifically, we stack outcomes for individual i in calendar quarter t across sub-experiments s, where each sub-experiment compares one early-initiation cohort to its matched cohort of initiators 16 quarters later:[15]

$$Y_{ist} = \alpha_i + \lambda_{s\tau} + \sum_{k=-8, k\neq -1}^{15} \beta_k \left(\mathbf{1}\{\tau_{ist} = k\} \times EarlyInitiator_{is} \right) + \varepsilon_{ist}. \tag{1}$$

where $EarlyInitiator_{is}$ is an indicator equal to one if patient i belongs to the early-initiation cohort in sub-experiment s, α_i are individual fixed effects, and $\lambda_{s\tau}$ are sub-experiment-by-event-time fixed effects. Event-time τ_{ist} is defined relative to the initiation quarter of the early-initiation cohort in sub-experiment s, so that the later-initiation cohort serves as the not-yet-treated control group throughout the estimation window. We follow outcomes from eight quarters before to fifteen quarters after initiation, omitting quarter -1 as the reference period. Heteroskedasticity-robust standard errors are clustered at the individual level.

Under the standard parallel trends assumption that later initiators would have evolved like early initiators in the time periods around the early initiators' initiation date, β_k coefficients estimate the causal impact of GLP-1 treatment on outcomes. Although the parallel trends assumption cannot be tested directly, pre-initiation coefficients allow us to assess its plausibility. Even after matching, early and later initiators may differ in unobserved disease progression, life cycle and labor market trajectories that affect the timing of initiation and subsequent outcomes.

Analysis Sample. Our analysis sample consists of all individuals in Denmark who initiate Ozempic for the first time between August 2018 and December 2019 or between August 2022 and December 2023. We restrict the sample to adults between the ages of 30 and 59 in December 2017 to ensure that all individuals are no older than 65 by the end of the post-period, and are thus more likely to be attached to the labor market.[16] Because later initiators necessarily survive the four-year follow-up period, we impose the same survival requirement on early initiators, yielding a balanced panel with fixed sample composition throughout follow-up.[17] Of the 9,794 patients who initiate Ozempic between August 2018 and December 2019, and of the 23,480 who initiate between August

[15] Appendix Table B.1 shows the early and later initiator pairs and counts for each sub-experiment.

[16] In Appendix Figure B.12 we exclude individuals born before 1960, who are eligible for early retirement by 2023.

[17] Before imposing this restriction, 98.0% of early initiators survive for four years after initiation. For reference, the corresponding four-year mortality risk for the average 50 year old male Dane is 1.2% (Statistics Denmark, 2025).

2022 and December 2023, our matching procedure successfully matches 7,021 of the early initiators (with a 71.7% match rate) to 7,021 later initiators.

Summary Statistics and Covariate Balance. Table 1 reports summary statistics for early and later Ozempic initiators before and after matching. Columns (1) and (2) show that, prior to matching, early and later initiators differ substantially along several dimensions. Early initiators are older, more likely to be male, and much more likely to have diabetes, consistent with prior evidence on Semaglutide adoption in Denmark (Mailhac et al., 2024). They also have lower annual income, lower employment, and higher rates of being out of the labor force than later initiators. 5–7% of initiators have a hospital-recorded obesity diagnosis; because this measure relies on hospital diagnoses it understates the prevalence of BMI-defined obesity. For context, 26.2% of Danes diagnosed with type 2 diabetes between 2011 and 2023 had a BMI above 35, compared with 37–42% of U.S. adults with type 2 diabetes at the same threshold (Kristensen et al., 2024; Boye et al., 2021).

Columns (3) and (4) show that matching substantially improves balance. By construction, matched early and later initiators are nearly identical in sex, age, diabetes status, and obesity status. The matched groups are also closely balanced on marital status, education, baseline annual income, labor force status, employment, and long-term sickness leave. In the matched sample, 55% are male, the average age is 50, and 84% had either a hospital diabetes diagnosis or a diabetes medication prescription filled in 2016–2017. Our sample is relatively socioeconomically disadvantaged; 85-87% have less than a four-year college-equivalent degree and annual total income is approximately DKK 292,000.

Identification in our stacked difference-in-differences design relies on the parallel trends assumption rather than on balance in baseline levels. Still, the improved covariate balance supports the use of later initiators as a comparison group. Section 5.3 presents robustness checks that assess the sensitivity of the estimates to alternative matching specifications and comparison groups.

5 Results

5.1 First Stage and Health Utilization

Appendix Figure B.2 shows the effects of Ozempic initiation on any Semaglutide and any Ozempic use and Table 2 presents the corresponding difference-in-differences estimates. By construction, the

coefficient in the quarter of initiation equals one, since treated patients begin treatment in that quarter while the comparison group consists of patients who initiate 16 quarters later. The coefficients on the subsequent quarters can be interpreted directly as treatment persistence (adherence). The figure shows high persistence in GLP-1 use. Although discontinuation is meaningful during the first year, adherence stabilizes markedly thereafter. From roughly quarter 6 onward, the share of treated patients remaining on GLP-1 treatment is relatively flat, with 75% still on treatment four years after initiation. This adherence rate is substantially higher than those reported in the U.S., where roughly half of type 2 diabetes patients discontinue after one year (Weiss et al., 2020).[18] This persistent and sustained treatment exposure implies that our design is well-powered to detect downstream effects on labor market outcomes.

Next, we investigate changes to health utilization-based outcomes around GLP-1 initiation. Appendix Figure B.4 shows a decline in cardiovascular drug use after an initial increase at treatment initiation, and a decline in emergency department encounters. These patterns are consistent with improved cardiometabolic health and with the clinical trial evidence on the health effects of GLP-1 treatment (Marso et al., 2016; Lincoff et al., 2023).

Both general practitioner services and cardiovascular drug use exhibit increases leading up to initiation and a spike at initiation. These increases may reflect the additional clinical contact involved in initiating treatment, including prescribing, medication review, and dose titration. More generally, these outcomes measure healthcare utilization, which can respond to care-seeking, monitoring, and prescribing behavior as well as to underlying health. We therefore interpret the patterns as suggestive context, rather than as direct evidence on the health mechanism linking GLP-1 initiation to downstream outcomes.

5.2 Long-Term Sickness Leave and Labor Market Outcomes

We begin by examining long-term sickness leave, the outcome most closely aligned with severe health-related limitations on work capacity. Panel (a) of Figure 1 shows that a statistically significant reduction in days of long-term sickness leave emerges by the end of the first year after initiation and becomes more pronounced in the second year and beyond. The estimated reduction is 0.809

[18] Differences in affordability may contribute, but so may differences in who uses the drug: most patients in our sample have type 2 diabetes, whereas much of U.S. GLP-1 use is for weight management. Consistent with this, adherence is somewhat lower among Danish Wegovy initiators, most of whom do not have diabetes (Appendix C).

days in the first two post-initiation years, and 0.912 days in years 3 and 4. Averaging over the entire four-year post period, the difference-in-differences estimate implies a statistically significant reduction of 0.864 days relative to a pre-period mean of 3.789 days, corresponding to a 22.8% decline (column 2 of Table 2). We see qualitatively similar reductions in other measures of sickness leave, including share of months with any long-term sickness leave, an indicator for any long-term sickness leave in a quarter, annual sickness leave benefits, and short-term sickness absence days (see Appendix Figure B.3 and Appendix Table B.2).

How large are these effects relative to the existing literature on pharmaceutical innovation and work capacity? The closest benchmark is Bütikofer and Skira (2018), who study the causal effects of Vioxx, a COX-2 inhibitor used to treat pain, on sickness-absence days beyond the first 16 employer-paid days in Norway. Our estimated 22.8% reduction is somewhat larger than the estimated effects of Vioxx, whose entry reduced sickness absence days among individuals with joint pain by 7–12%, while its withdrawal increased them by 12–16%. The implied effects in our setting are larger in absolute terms as well, as long-term sickness leave is more common (7.0–7.7% of person-quarters, compared with 3–4% in their sample). The larger reduction may reflect that GLP-1s improve functioning along several margins at once, whereas Vioxx targets pain.

An important question is whether this reduction in long-term sickness leave translates into broader labor market outcomes. Panels (b)–(d) of Figure 1 show the effects on employment, labor force participation, and log total income. Across these outcomes, we find no statistically significant or economically meaningful effects. The two-period difference-in-differences estimate for being employed is -0.0049 (SE = 0.0050). Given a baseline mean of 64.9%, this allows us to rule out employment reductions of more than 1.5 percentage points at the 95% confidence level. Similarly, the estimate for labor force participation is -0.0019 (SE = 0.0047), relative to a baseline mean of 67.7%, allowing us to rule out gains in labor force participation of more than 0.7 percentage points. For log total income, the estimate is -0.0010 (SE = 0.0084), implying that we can rule out income increases exceeding 1.6%.

The null result for total income may nevertheless mask changes in its individual components. Appendix Figure B.5 decomposes total income into its different sources. While the post-initiation estimates suggest lower wage income and higher non-sickness-leave transfers—the two largest components—the magnitudes are modest (less than 4,000 DKK) and both event studies

exhibit meaningful pre-treatment differences. We therefore interpret the component estimates descriptively rather than causally.

More generally, matching improves the comparability of early and later initiators on observed baseline characteristics, but it does not ensure parallel pre-treatment paths for every outcome. We therefore treat analyses of secondary outcomes as supplementary evidence and place the greatest weight on the primary outcomes, whose event studies have stable pre-treatment paths. We turn next to the robustness of the main results.

5.3 Robustness

We assess the robustness of our primary outcomes using several alternative specifications and estimators. The results of these exercises can be found in Appendix Figure B.6.

First, we vary the timing of covariate measurement in the matching procedure. Our baseline specification matches patients using fixed pre-period characteristics measured in 2016–2017. As an alternative, we match on characteristics measured in the year prior to each subexperiment's own treatment initiation. For example, 2019 initiators are matched with 2023 initiators on diabetes and obesity status in 2018. All other features of the matching procedure are unchanged. This exercise assesses whether the results are sensitive to matching on initiation cohorts' immediate pre-initiation clinical state, which aligns treatment need at the time of initiation, rather than on fixed baseline characteristics.

Second, we consider alternative comparison group constructions motivated by the concern that patients initiating four years apart may not be the ideal counterfactual. Specifically, we implement the Callaway and Sant'Anna (2021) estimator using later-initiation patients as controls. Relative to our baseline stacked design, which fixes the comparison group to patients initiating 16 quarters later, this approach uses all later-treated cohorts for each cohort-time comparison, yielding a substantially larger estimation sample of 59,443 initiators. Consequently, the longest-horizon estimates are identified from a sample very similar to the baseline design, while the earlier post-treatment coefficients draw on a broader set of later initiators that initiate closer in time to the early initiators and may therefore provide a more credible counterfactual. In this sense, the main advantage of this estimator is that it shortens the treatment-control gap, while also providing a useful robustness check for the short- and medium-run dynamics and, as expected, increasing precision at shorter

horizons.

Third, we restrict the sample to the first four initiation cohorts, from 2018Q3 through 2019Q2, excluding the final two early-treatment quarters whose matched controls initiate in the second half of 2023 when high uptake of Wegovy, a potential substitute for Ozempic, may have altered selection into Ozempic initiation. A further advantage of this restriction is that it increases the match rate to 87%.

Across all three alternative specifications and estimators, our main findings are robust. Ozempic initiation significantly reduces long-term sickness leave by about 0.45–0.75 days per quarter, with the effects peaking during the second year. We find no significant aggregate effects on employment, labor force status, or log total income across robustness checks.

Finally, we examine whether our findings for Ozempic initiation extend to Wegovy, the Semaglutide formulation approved for weight management and a rapidly expanding treatment margin. In Appendix C, we follow the design of Diamond (2026) and restrict the analysis to quarterly outcomes, as the short window of observed Wegovy initiations limits the availability of later-initiating controls for annual outcomes. We find a reduction of similar magnitude in long-term sickness leave over the first three post-initiation quarters.

5.4 Heterogeneity

We examine heterogeneity along a few relevant dimensions: demographic and socioeconomic characteristics (age, sex, and educational attainment); baseline labor force status, given our focus on labor market outcomes; and underlying health (prior cardiovascular medication use and hospital contact), since clinical trials show that health impacts differ by baseline health status.

Reductions in long-term sickness leave are apparent across all subgroups and are statistically significant in most cases. The decline is largest among initiators with prior hospitalizations, the group with the highest baseline level of sickness leave. For labor market outcomes, no subgroup shows a statistically significant gain. The 95% confidence intervals rule out increases in employment or labor force participation of more than 2 percentage points in almost every group. The only statistically significant estimates are small declines in employment for men, those without higher ed-

ucation, and those in the labor force at baseline.[19] We interpret these subgroup patterns cautiously, as the event studies are relatively noisy and some subgroups exhibit diverging pre-trends.

Finally, we examine treatment effects among women who were not employed at baseline, a group for whom Diamond (2026) finds large increases in employment in the United States. We do not detect statistically significant changes in labor market outcomes and can rule out effects larger than a few percentage points (Appendix Figure B.11).[20] Because this subsample is defined by baseline non-employment, the pre-treatment coefficients for employment provide limited information: the pre-period coefficients are mechanically zero for women who remain non-employed throughout the pre-period and are thus identified off of early and late initiating women who were previously employed but became non-employed in 2017.

5.5 Economic and Fiscal Benefits of GLP-1s

Finally, we provide a back-of-the-envelope assessment of the economic and fiscal benefits associated with reduced sickness absence, with detailed calculations reported in Appendix D. Valuing the estimated reduction in long-term sickness-leave days at average daily earnings yields an annual benefit of DKK 4,437 per employed initiator (approximately USD 672), equivalent to 1.5% of annual labor earnings. Assuming that the estimated 22.8% reduction in long-term sickness leave days also applies to short-term absences raises this value to DKK 6,557 (approximately USD 993), or 2.2% of annual labor earnings. These figures represent the total wage-valued gains from reduced absence to the employer, municipality, and individual. Applied to the average annual income of an American adult with diabetes, these percentages imply annual benefits of approximately USD 826 to 1,221.

We also estimate an annual reduction in municipality-paid sickness benefits to the individual of DKK 545 per initiator, equivalent to DKK 839 when normalized by the employment share. This estimate is not statistically significant and provides imprecise evidence of a potential public budgetary offset to GLP-1 subsidies. Under the assumptions of full wage replacement through-

[19] Among men, long-term sickness leave rises slightly before initiation (Appendix Figure B.7), which suggests that early-initiating men may be on a worsening health trajectory that is not captured by our matching procedure. Among initiators without higher education, employment starts to fall before initiation (Appendix Figure B.8), so the subsequent decline may reflect a pre-existing trend.

[20] These confidence intervals lie well below the employment gains reported by Diamond (2026) for American women initiating GLP-1 treatment primarily for weight loss, a setting that differs from ours both institutionally and in sample composition. Her proposed mechanism operates in part through the obesity penalty, and the relevance of channels such as discrimination, productivity, and health may differ across the U.S. and Danish contexts and across the two study samples.

out sickness spells and complete measurement of municipal savings, the implied allocation of the combined sickness leave benefit is 12.8% to municipalities and 87.2% to employers.

These estimates may not generalize directly to other institutional settings. In countries with less generous sickness leave systems, such as the United States, the direct fiscal savings we quantify will likely be smaller, since less of the cost of long-term absence is currently borne by employers or the public sector. The corresponding gains may instead show up as increased productivity, less presenteeism, and a lower risk that health shocks disrupt employment altogether. Even with this caveat, the estimated savings for Denmark are a real budgetary offset to the cost of GLP-1 treatment: they imply that the public cost of subsidizing these drugs is lower than the pharmacy cost alone would suggest.[21]

6 Conclusion

This paper provides new evidence on the economic consequences of GLP-1 treatment using linked Danish administrative data and a matched stacked difference-in-differences design. Comparing early Ozempic initiators to observably similar patients who initiate four years later, we find that GLP-1 treatment leads to a meaningful and persistent reduction in long-term sickness leave. This effect grows over time and is accompanied by declines in emergency department encounters and cardiovascular medication use, consistent with sustained treatment exposure and gradual improvements in underlying health. However, this potential increase in work capacity does not translate into improved labor market outcomes over four years: we estimate precise null impacts on employment, labor force participation, and total income. Even in the absence of detectable labor market gains, GLP-1 treatment generates meaningful fiscal spillovers by reducing prolonged sickness absence, valued at 1.5–2.2% of annual labor income per employed individual.

These findings speak to ongoing policy debates over GLP-1 coverage, but several institutional features may shape how the magnitudes translate across settings. Early Danish Ozempic adopters are likely more clinically indicated than the marginal patients who would initiate under broader coverage expansions, so our estimates are most informative about gains among patients with clear medical need. Institutional context also matters: in the United States, for example, GLP-1 avail-

[21] While negotiated net prices may differ from patient prices, these fiscal benefits remain substantially smaller than the current cost of GLP-1 treatments in 2026.

ability, sickness benefit systems, and public financing all differ from Denmark's, so the fiscal gains we identify through reduced long-term sickness leave need not map one-for-one into U.S. budget savings.

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Tables and Figures

Table 1: Summary Statistics

	All Early Initiators (1)	All Late Initiators (2)	Matched Early Initiators (3)	Matched Late Initiators (4)	Absolute Std. Difference (5)
Patient Characteristics:					
Male	0.560	0.461	0.545	0.545	< 0.001
Diabetic	0.884	0.293	0.839	0.839	< 0.001
Obese	0.073	0.052	0.058	0.058	< 0.001
Age	50.201	47.693	49.931	49.907	0.003
	(6.869)	(7.676)	(6.985)	(7.010)	
Married	0.531	0.544	0.521	0.545	0.047
Unmarried	0.276	0.262	0.284	0.267	0.038
Divorced	0.177	0.182	0.176	0.173	0.006
Widowed	0.016	0.011	0.019	0.015	0.033
No Higher Education	0.366	0.302	0.391	0.364	0.056
Some Higher Education	0.451	0.457	0.443	0.454	0.021
More Higher Education	0.154	0.222	0.130	0.150	0.058
Outcomes at Baseline:					
Long-Term Sickness Leave, in Days	3.756	3.549	3.981	3.353	0.048
	(13.166)	(12.377)	(13.761)	(12.294)	
Any Long-Term Sickness Leave	0.076	0.074	0.077	0.070	0.029
Employed	0.666	0.747	0.634	0.665	0.065
In Labor Force	0.695	0.777	0.660	0.697	0.081
Annual Total Income, in DKK	302,117	331,094	290,829	293,414	0.015
	(177,030)	(192,775)	(169,195)	(167,255)	
Observations	9,794	23,480	7,021	7,021	

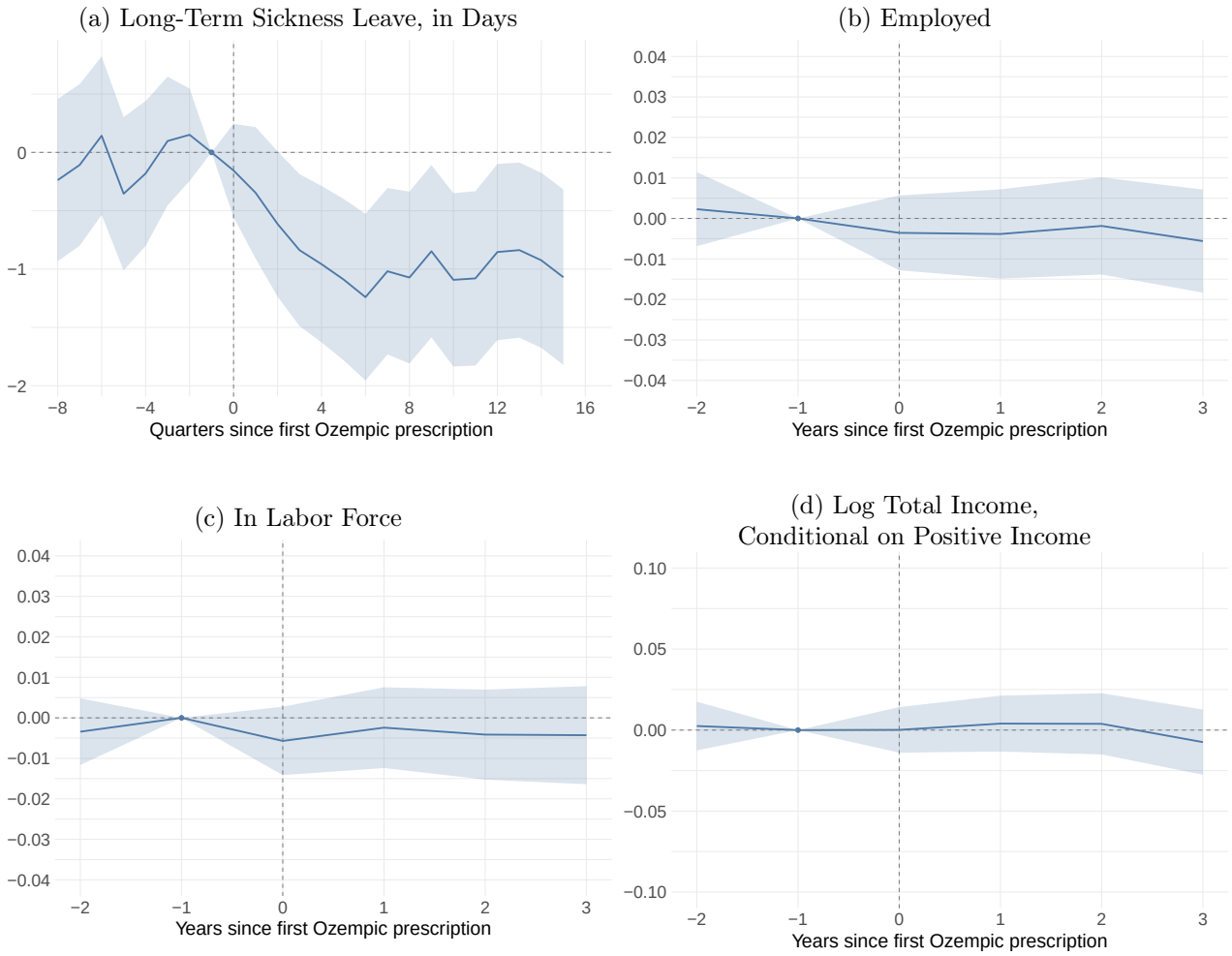
Notes: This table presents summary statistics and covariate balance. Column (1) displays all early GLP-1 initiators (August 2018 – December 2019), column (2) displays all late GLP-1 initiators (August 2022 – December 2023); see Appendix Table B.1 for initiator counts by quarter. Columns (3) and (4) display the subset of matched early and later initiators. Column (5) displays the absolute standardized differences (between columns 3 and 4). Diabetic status is defined as being on a diabetes medication or having a hospital diabetes diagnosis in 2016–2017 and obesity status is defined as having a hospital obesity diagnosis in 2016–2017. All other variables are measured in 2017; monetary amounts are expressed in 2025 DKK. No, some, and more higher education are defined as anything equal to or less than upper secondary education; vocational training, short-cycle higher education; and vocational bachelors, bachelors, masters, PhD, respectively. Long-term sickness leave in days and any long-term sickness leave are measured quarterly, while employed, in labor force, and annual total income are measured yearly. Standard deviations are reported in parentheses below the means for non-indicator variables.

Table 2: Difference-in-Differences Estimates

	Dependent Variable:				
	Semaglutide	Sickness Leave, in Days	Employed	In Labor Force	Log Total Income
	(1)	(2)	(3)	(4)	(5)
Aggregate Over Four Years:					
EarlyInitiator	0.777***	−0.864***	−0.005	−0.002	−0.001
	(0.004)	(0.220)	(0.005)	(0.005)	(0.008)
Years 1–2:					
EarlyInitiator	0.807***	−0.809***	−0.005	−0.001	0.003
	(0.004)	(0.234)	(0.005)	(0.005)	(0.009)
Years 3–4:					
EarlyInitiator	0.751***	−0.912***	−0.005	−0.003	−0.003
	(0.004)	(0.259)	(0.006)	(0.005)	(0.009)
Mean Dep. Var	0	3.789	0.649	0.677	12.387
Observations	322,966	322,966	70,185	70,185	69,929

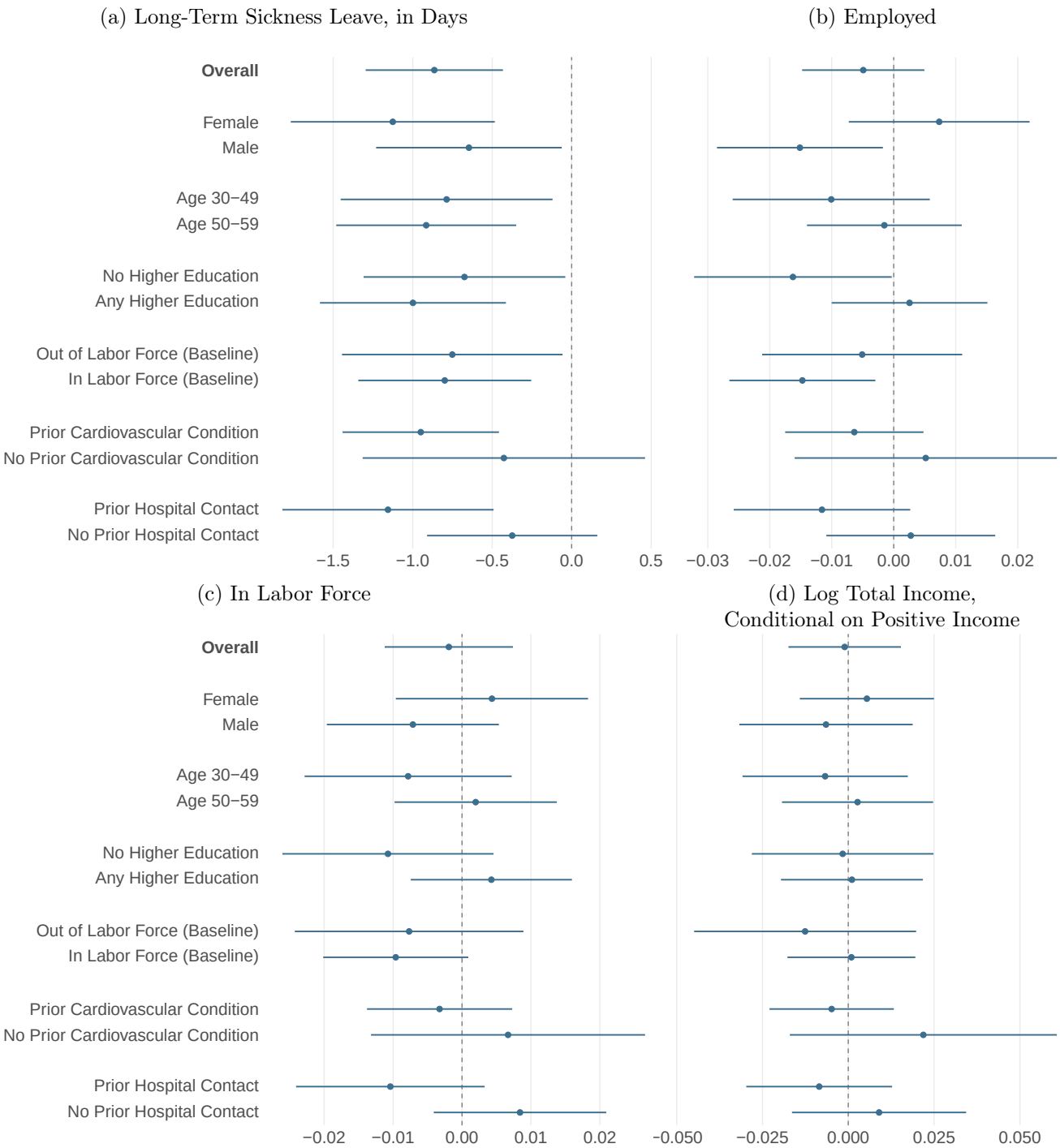
Notes: This table presents the aggregate difference-in-differences estimates that correspond to the event study specification in Equation (1). Long-term sickness leave is measured in days conditional on lasting more than 30 days. Semaglutide and sickness leave are measured quarterly while employed, in labor force, and annual total income are measured yearly. The top row displays the aggregate over the four-year post-period vs two-year pre-period. The middle row displays the aggregate over the first two-year post-period vs two-year pre-period (omitting event time zero). The bottom row displays the aggregate over the second two-year period (years 3–4) vs two-year pre-period. Means for the dependent variable correspond to the average over the pre-period. Log total income (in DKK) is conditional on positive total income (99.6% of our patient-year observations have positive total income). The sample is restricted to individuals who survive throughout the four-year follow-up period. Heteroskedastic standard errors clustered at the individual-level are displayed in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 1: Primary Outcomes



Notes: This figure plots the estimated event study coefficients from Equation (1) for days of long-term sickness leave (lasting more than 30 days), indicator for being employed, indicator for being in the labor force, and log total income (in DKK) conditional on positive income (99.6% of our patient-year observations have positive income) from panels (a) to (d). Long-term sickness leave is measured quarterly while the other outcomes are yearly. The sample consists of early Ozempic initiators (August 2018 to December 2019) matched to later initiators (August 2022 to December 2023). Patients are one-to-one exact-matched on sex, five-year age bins, diabetes, and obesity status in 2016–2017 and propensity score matched within strata. The sample size consists of 14,042 patients, of which exactly half are early initiators. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. Baseline means over the pre-period are 3.789, 0.649, 0.677, and 12.387 in log (mean total income is DKK 286,430), for panels (a) to (d), respectively.

Figure 2: Heterogeneity Analysis – Difference-in-Differences Estimates



Notes: This figure plots difference-in-differences estimates over four years for our primary outcomes for subsamples corresponding to six margins of heterogeneity: sex, age, educational attainment, labor force status, cardiovascular medication use, and hospitalization status at baseline. Appendix Figures B.7 to B.10 show plots of the event study coefficients. The number of early initiators for each subsample going from top to bottom (including overall) are: 7,021, 3,193, 3,828, 2,782, 4,239, 2,997, 4,024, 2,389, 4,632, 5,765, 1,256, 3,846, 3,175. These heterogeneity subsamples are created after performing the overall matching procedure. The estimates are qualitatively similar when matching after creating separate heterogeneity subsamples. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Online Appendix for "The Labor Market Impacts of GLP-1s"

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A Data Sources

This appendix describes in more detail the Danish administrative registers used in this paper. All registers cover the entire population, can be linked at the individual level using unique personal identifiers, and span the period 2016–2023. Appendix Table A.1 details the construction of the variables used in the analyses.

GLP-1 prescriptions. We identify the date of treatment initiation and construct the indicator for any Semaglutide prescription fill from the *National Prescription Register* (*LMDB*). This register includes the universe of filled prescriptions in Denmark, recording each drug's ATC code, the exact date when the prescription was filled, and the quantity dispensed. The ATC code classifies drugs by active chemical substance and therapeutic use, allowing us to distinguish products such as Ozempic and Wegovy.

Long-term sickness leave. Based on information from the *Danish Register for Evaluation of Marginalization* (*DREAM*), we construct several measures of long-term sickness absence. Long-term sickness leave absences are reported at the weekly-level, which we convert to days in a quarter by multiplying each indicator by seven and summing across weeks in a quarter. Stapelfeldt et al. (2012) find that the *DREAM* register identifies long-term sickness-absence spells with high accuracy, although spell duration is overestimated by an average of 1.4 weeks relative to workplace records.

Second, we construct the share of months in each quarter with any long-term sickness absence. Third, we construct an indicator for whether any long-term sickness absence occurs within the quarter. We obtain annual municipality-paid sickness leave benefits from the *Income Register* (*IND*), which provides annual information extracted from the tax records of every Danish resident.

Labor market outcomes. We construct the indicators for employment and for being out of the labor force based on the information in the *Register-Based Labor Force Statistics* (*RAS*). This register provides an annual snapshot of each Danish resident's labor market status at the end of November and is used by the Danish government to produce official labor market statistics. We obtain annual total income from the Income Register defined above.

Healthcare utilization. We construct the indicator for any emergency department encounter based on the information in the *National Patient Register* (*LPR*), which records all inpatient, outpatient, and emergency hospital contacts in public hospitals together with their diagnosis codes and the exact date of the contact. We use the National Prescription Register defined above to create the indicator for any cardiovascular drug prescription. We obtain the number of primary care services received from the *Health Insurance Register* (*SSSY*). This register consists of reimbursement claims submitted by general practitioners (GPs) and other specialists who contract with the national health insurance and operate outside of hospitals. Each claim includes the date of the contact, for both in-person and electronic contacts, and the services delivered during consultations. The *SSSY* unfortunately does not include diagnosis codes and none of our health data sets include biomarkers such as weight or blood glucose levels.

Baseline covariates. We construct the indicators for type 2 diabetes and obesity based on the diagnoses from the *National Patient Register*; for type 2 diabetes, we additionally use information on prescription fills for diabetes medication from the *National Prescription Register*. From the *National Patient Register*, we also obtain the annual sum of hospital expenditures, covering inpatient,

outpatient, and emergency room contacts. We obtain marital status and region of residence from the *Population Register* (*BEF*), which records demographic and household characteristics for every Danish resident on December 31 of each year. We obtain the highest educational attainment from the *Education Register* (*UDDA*), which details the highest completed qualification for every Danish resident as of September 30 of each year. Occupation and industry codes come from *RAS*. Finally, we obtain the number of GP services from the *Health Insurance Register*.

Short-term sickness leave. We construct the number of days of absence from the *Absence from Work* data set (*FRPE*). This data set records the start and end dates of absence spells by cause, including own illness, child illness, and maternity or adoption leave. We restrict our attention to spells lasting less than 30 days due to own illness. The data set covers the entire public sector, all private firms with more than 250 full-time employees, and a size- and industry-stratified sample of smaller private firms with at least 10 full-time employees that changes from year to year. For this reason, we restrict our attention to employees in the firms that are always included in the data set in every year (*i.e.*, publicly-owned firms and privately-owned firms with more than 250 employees).

Appendix Table A.1: Data Sources

Variable	Definition	Registry name
Prescriptions and Diagnoses		
GLP-1 (Semaglutide)	Indicator for filled prescriptions of Semaglutide (ATC: A10BJ06); we also create a separate indicator for Ozempic from the brand name	National Prescription Register
Diabetes drugs	Indicator for filled prescriptions of diabetes drugs (ATC: beginning with 'A10')	National Prescription Register
Cardiovascular drugs	Indicator for any cardiovascular drug prescription in a given quarter (ATC: beginning with 'C')	National Prescription Register
Type 2 diabetes	Indicator based on hospital diagnosis codes (ICD-10: E11) or filled prescriptions for diabetes medication (ATC: beginning with 'A10')	National Patient Register; National Prescription Register
Obesity	Indicator based on hospital diagnosis codes (ICD-10: E66, excluding E66.3 overweight)	National Patient Register
Healthcare Utilization		
General Practitioner Services	Number of services received from the patient's general practitioner in a given quarter	Health Insurance Register
Emergency department encounter	Indicator for an emergency department hospital encounter in a given quarter	National Patient Register
Hospital expenditures	Annual sum of expenditures on hospital contacts, including inpatient, outpatient, and emergency room contacts, expressed in 2025 DKK.	National Patient Register
Labor Market Variables		
Employment status	Indicators for being out of the labor force or employed as of November of that year	Register-Based Labor Force Statistics
Long-term sickness leave (days)	Number of days in a quarter with sickness leave, conditional on lasting longer than 30 days	DREAM
Long-term sickness leave (share)	Share of months in a quarter with sickness leave lasting more than 30 days	DREAM
Sickness leave benefits	Annual municipality-paid sickness leave benefits.	Income Register
Total income	Annual sum of wage earnings, self-employment income, sickness benefits, and other public transfers, expressed in 2025 DKK.	Income Register
Demographics and Covariates		
Age	Constructed from date of birth	Population Register
Marital/partnership status	Civil status, used to construct marital and partnership status	Population Register
Region of residence	From municipality of residence	Population Register
Educational attainment	Highest educational attainment, grouped into "no higher education" (equal to or less than upper secondary), "some higher education" (vocational training, short-cycle higher education), and "more higher education" (vocational bachelors, bachelors, masters, PhD)	Education Register
Industry	Industry based on the two-digit Danish Industry Classification (DB), corresponding to the EU NACE classification.	Register-Based Labor Force Statistics
Occupation	Occupation classification based on the two-digit Danish Classification of Occupations (DISCO-08), the Danish version of the International Standard Classification of Occupations (ISCO-08).	Register-Based Labor Force Statistics

Notes: This table details the data sources used in this paper. All data sources are available for years 2016–2023, except for the hospital expenditures variable which is available for years 2016–2017.

B Appendix Tables and Figures

Appendix Table B.1: Illustration of Sub-Experiments: Ozempic Initiators Over Time

Cohort ("Sub-experiment")	Early Initiators	Later Initiators	Matched
2018 Q3 / 2022 Q3	552	4,946	547
2018 Q4 / 2022 Q4	1,885	5,597	1,483
2019 Q1 / 2023 Q1	1,918	6,378	1,833
2019 Q2 / 2023 Q2	1,855	3,990	1,552
2019 Q3 / 2023 Q3	1,605	1,057	678
2019 Q4 / 2023 Q4	1,979	1,512	928
Total	9,794	23,480	7,021

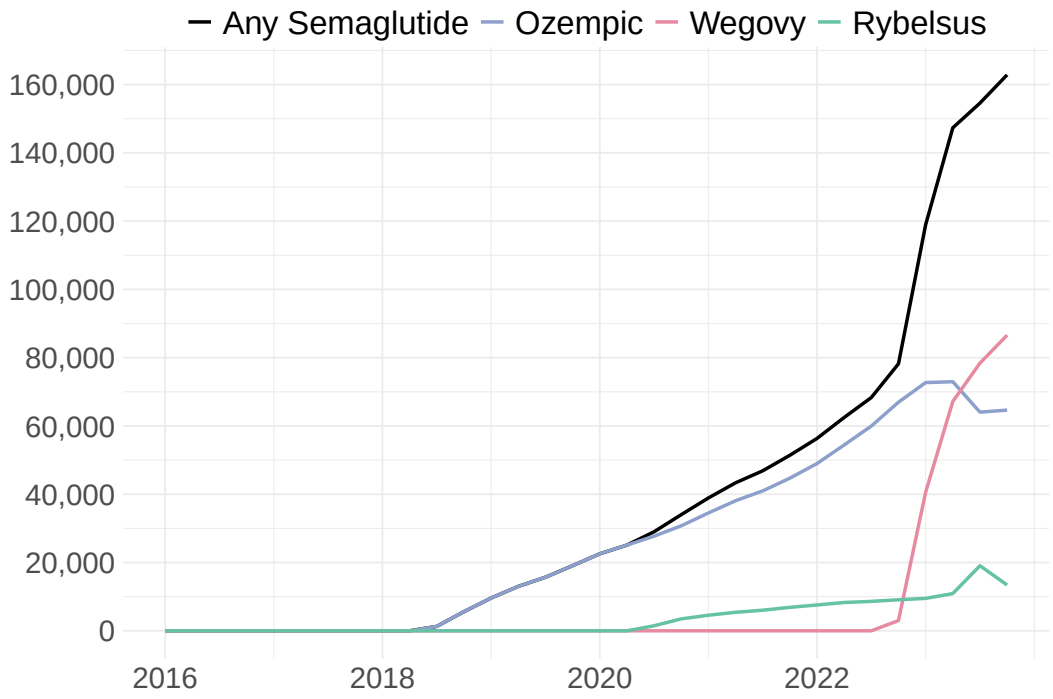
Notes: This table illustrates the sub-experiment cohorts of early versus later Ozempic initiators described in Section 4. The six rows correspond to the six sub-experiment cohorts. Row 1 shows that there are 552 (early) Ozempic initiators in the 2018 Q3. We pair these early initiators with initiators who begin Ozempic 16 quarters later in 2022 Q3, of which there are 4,946 individuals. Our matching procedure matches 547 later initiators to 547 early initiators, forming a first sub-experiment. We match 7,021 out of 9,794 early initiators (71.7%). Our final analysis sample size consists of 7,021 early and 7,021 later initiators (14,042 total).

Appendix Table B.2: Difference-in-Differences Estimates for Additional Outcomes

	Dependent Variable:							
	Ozempic	Sick Leave, Any	Sick Leave, Share	Sick Leave Benefits	Absent Days	Cardio. Drug, Any	GP Services	ER Encounter, Any
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Aggregate Over Four Years:								
EarlyInitiator	0.765***	−0.014***	−0.011***	−545.3	−0.318**	−0.007	−0.210***	−0.007***
	(0.004)	(0.003)	(0.003)	(374.2)	(0.154)	(0.004)	(0.044)	(0.002)
Years 1–2:								
EarlyInitiator	0.804***	−0.012***	−0.010***	−181.3	−0.386**	0.008**	0.064	−0.007***
	(0.004)	(0.004)	(0.003)	(408.1)	(0.171)	(0.004)	(0.046)	(0.002)
Years 3–4:								
EarlyInitiator	0.731***	−0.015***	−0.012***	−727.2*	−0.251	−0.020***	−0.450***	−0.007***
	(0.004)	(0.004)	(0.003)	(413.0)	(0.189)	(0.005)	(0.051)	(0.003)
Mean Dep. Var	0	0.077	0.054	3386.7	2.784	0.640	5.272	0.080
Observations	322,966	322,966	322,966	70,208	93,752	322,966	322,966	322,966

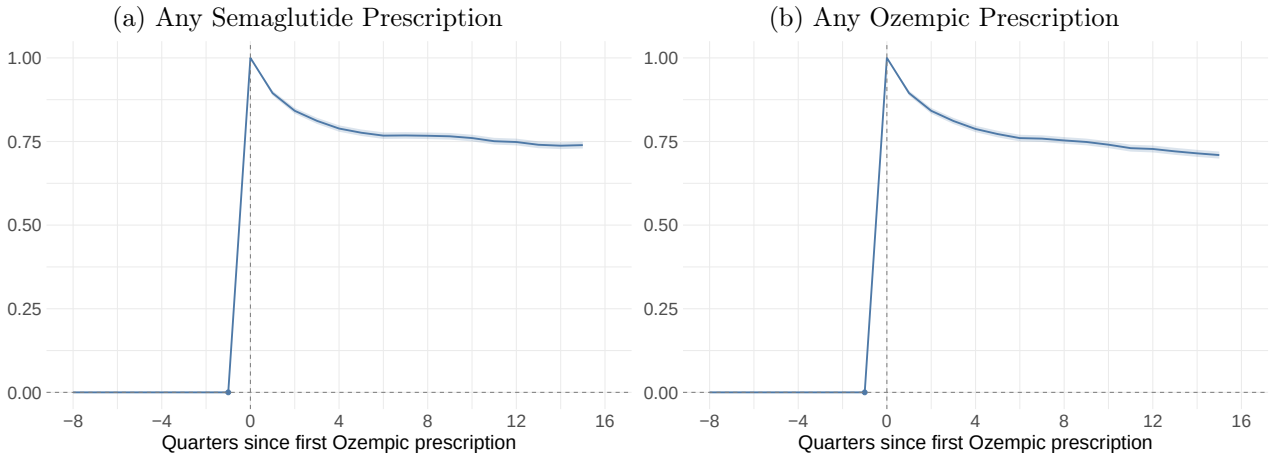
Notes: This table presents the aggregate difference-in-differences estimates that correspond to the event study specification in Equation (1) for additional outcomes not in Table 2. "Sick Leave, Any" corresponds to any month in a quarter with long-term sickness leave. "Sick Leave, Share" is the fraction of months in a quarter with long-term sickness leave. "Sick leave benefits" are yearly benefits paid to the individual by the municipality. "Absent Days" are the number of calendar days an individual is absent due to sickness in a calendar quarter. They include all short-term sickness absence lengths and are only available for individuals employed in the public sector and firms with more than 250 employees. The remaining columns report an indicator for any cardiovascular medication use, number of general practitioner services, and any emergency room encounter. All outcomes except sick leave benefits (annual) are measured at the quarterly-level. The top row displays the aggregate over the four-year post-period vs two-year pre-period. The middle row displays the aggregate over the first two-year post-period vs two-year pre-period (omitting event time zero). The bottom row displays the aggregate over the second two-year period (years 3–4) vs two-year pre-period. Means for the dependent variable correspond to the average over the pre-period. The sample is restricted to individuals who survive throughout the four-year follow-up period. Standard errors clustered at the individual-level are displayed. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Appendix Figure B.1: Time Series of Counts of Semaglutide Users, 2016–2023



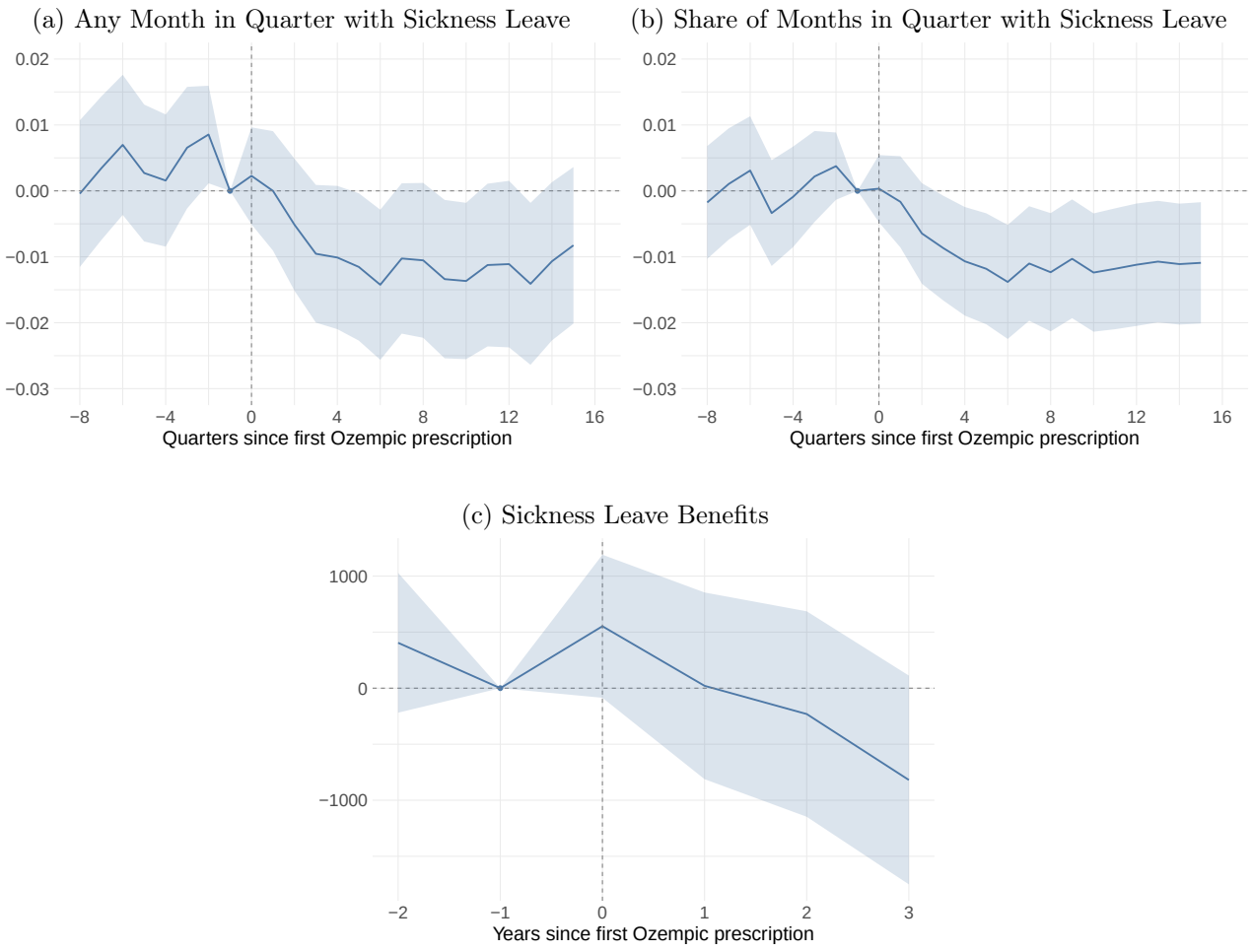
Notes: This figure presents the number of all unique individuals filling a Semaglutide prescription per quarter in Denmark, by type of Semaglutide from January 2016 to December 2023.

Appendix Figure B.2: Semaglutide and Ozempic Prescriptions



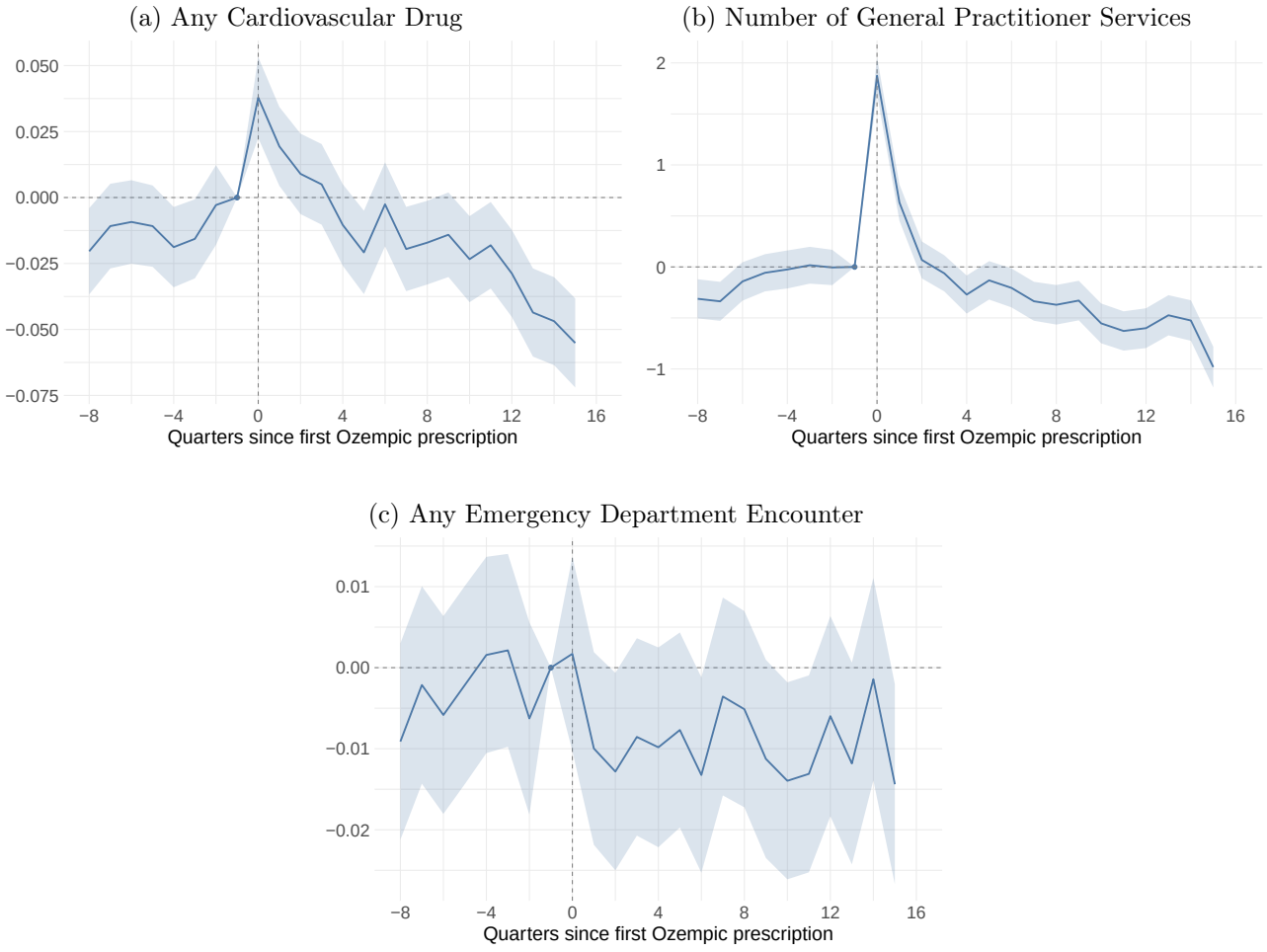
Notes: This figure plots the estimated event study coefficients from Equation (1) for any Semaglutide prescription (panel a) and any Ozempic prescription (panel b) in a given quarter. The sample consists of early Ozempic initiators (August 2018 to December 2019) matched to later initiators (August 2022 to December 2023). Patients are one-to-one exact-matched on sex, five-year age bins, diabetes, and obesity status in 2016–2017 and propensity score matched within strata. The sample size consists of 14,042 patients, of which exactly half are early initiators. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Appendix Figure B.3: Other Measures of Sickness Leave



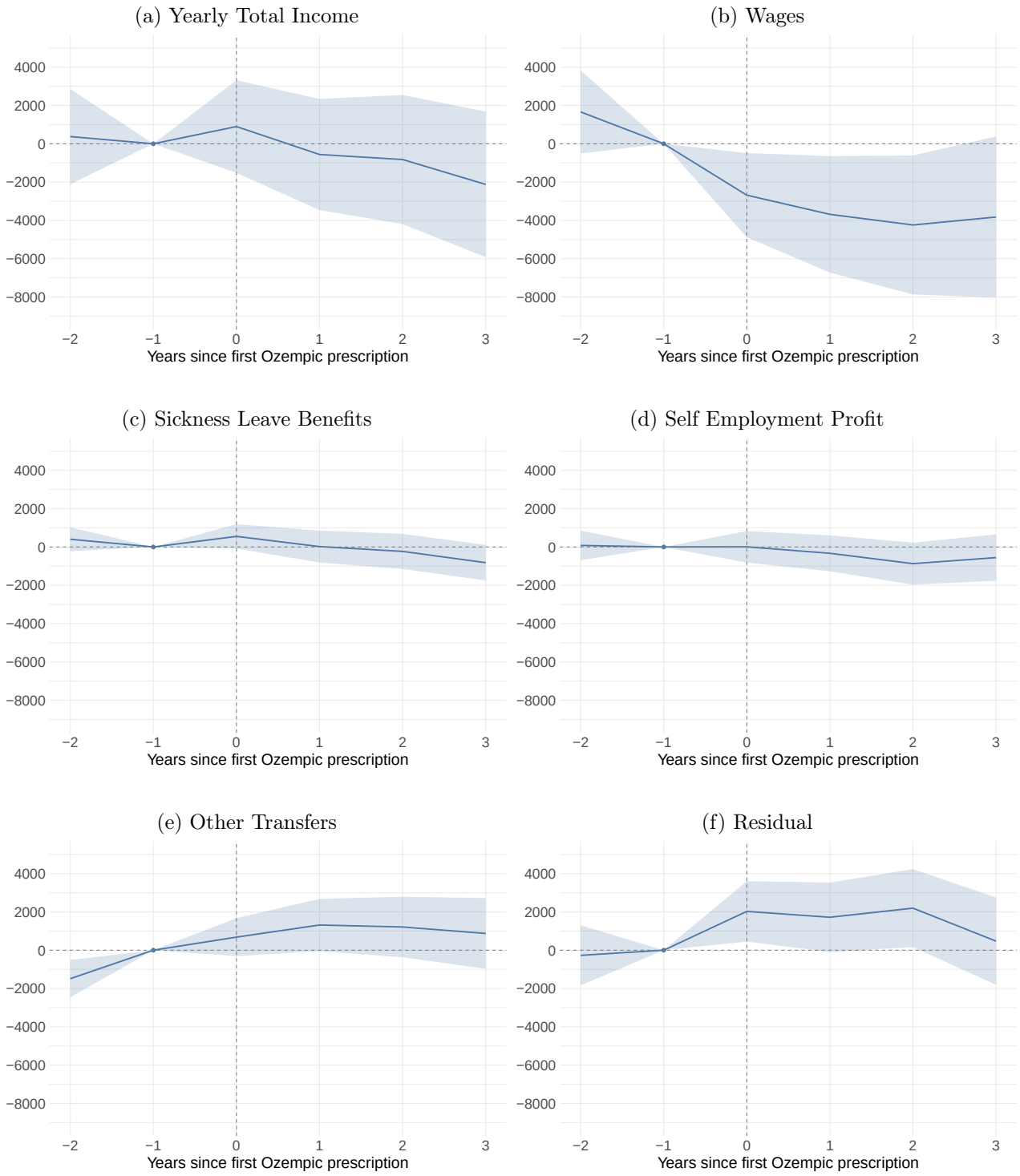
Notes: This figure plots the estimated event study coefficients from Equation (1) for three additional measures of sickness leave: any month in a quarter with long-term sickness leave, share of months in a quarter with long-term sickness leave, and annual total sickness leave benefits in DKK. The sample consists of early Ozempic initiators (August 2018 to December 2019) matched to later initiators (August 2022 to December 2023). Patients are one-to-one exact-matched on sex, five-year age bins, diabetes, and obesity status in 2016–2017 and propensity score matched within strata. The sample size consists of 14,042 patients, of which exactly half are early initiators. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. Baseline means over the pre-period are 0.077, 0.054, and 3387.

Appendix Figure B.4: Health-Related Outcomes and Utilization



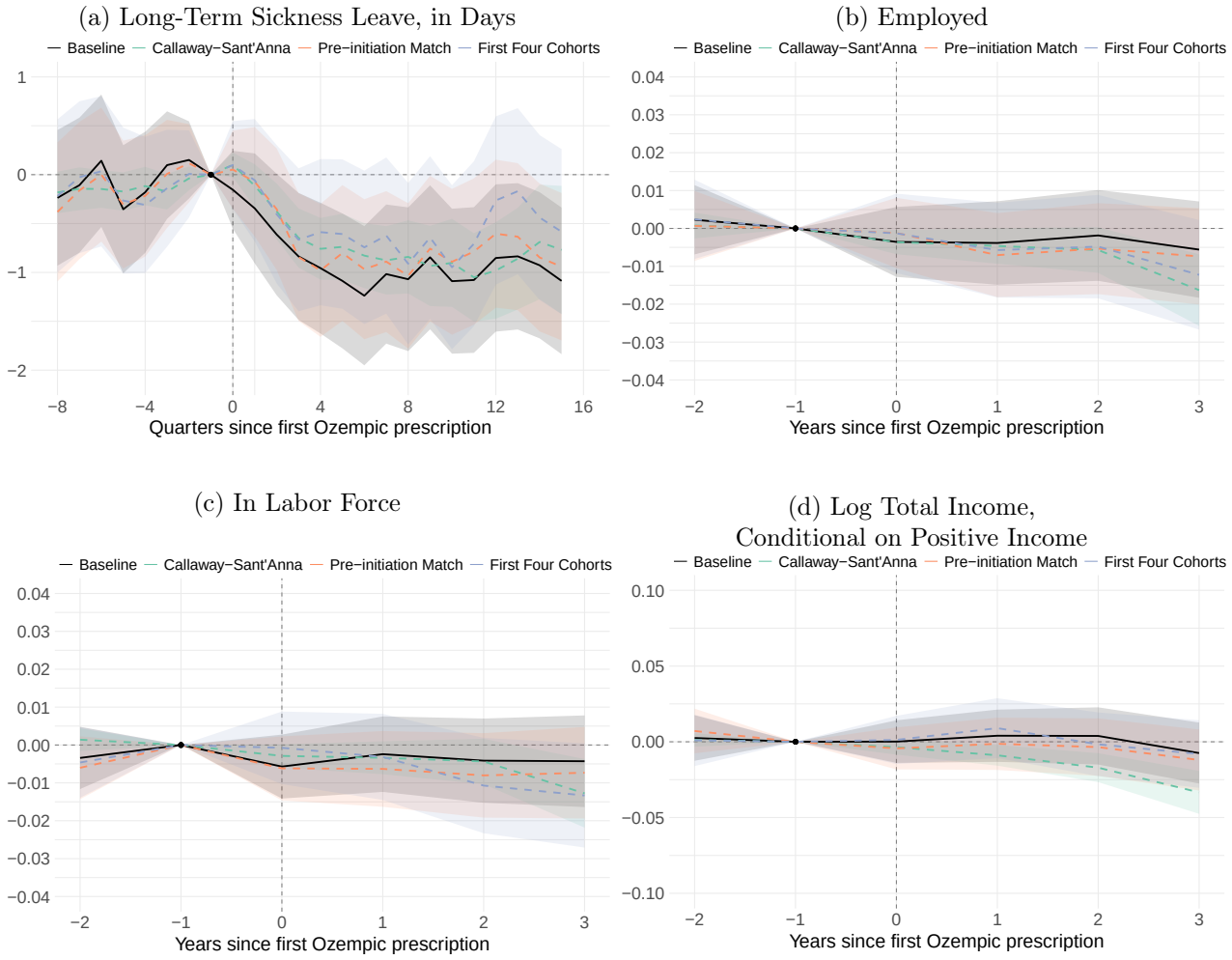
Notes: This figure plots the estimated event study coefficients from Equation (1) for health-related outcomes: any cardiovascular drug prescription, number of general practitioner services, and any emergency department encounter in a given quarter. Cardiovascular drugs are identified via ATC code 'C' which includes all drugs that target the cardiovascular system including antihypertensives, diuretics, beta blockers, calcium channel blockers, renin-angiotensin system agents, and lipid-lowering drugs. The sample consists of early Ozempic initiators (August 2018 to December 2019) matched to later initiators (August 2022 to December 2023). Patients are one-to-one exact-matched on sex, five-year age bins, diabetes, and obesity status in 2016–2017 and propensity score matched within strata. The sample size consists of 14,042 patients, of which exactly half are early initiators. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. Baseline means over the pre-period are 0.640, 5.272, and 0.080.

Appendix Figure B.5: Components of Yearly Total Income



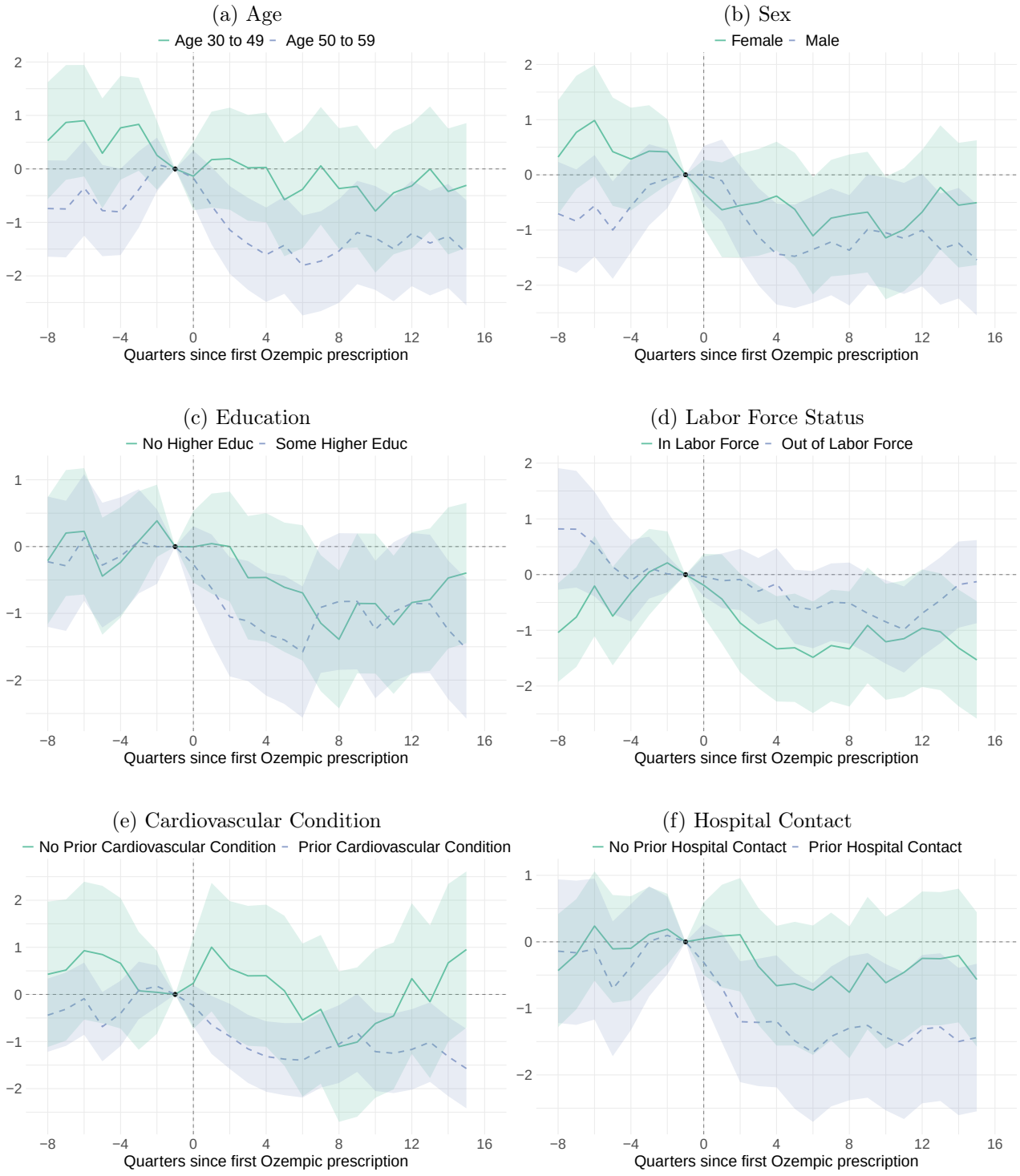
Notes: This figure plots the estimated event study coefficients from Equation (1) for yearly total income (panel a) and all components that sum to total income (panels b to f): wages, sickness leave benefits, self-employment profit, other transfers, and residual income (*e.g.*, private pension payouts, wealth income, and honorarium income). Note that sickness leave benefits in panel (c) are the same as in panel (c) of Appendix Figure B.3. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. Baseline means over the pre-period are 286,430, 196,382, 3,387, 8,874, 60,803, 16,985 for panels (a) to (f) respectively.

Appendix Figure B.6: Robustness of Primary Outcomes



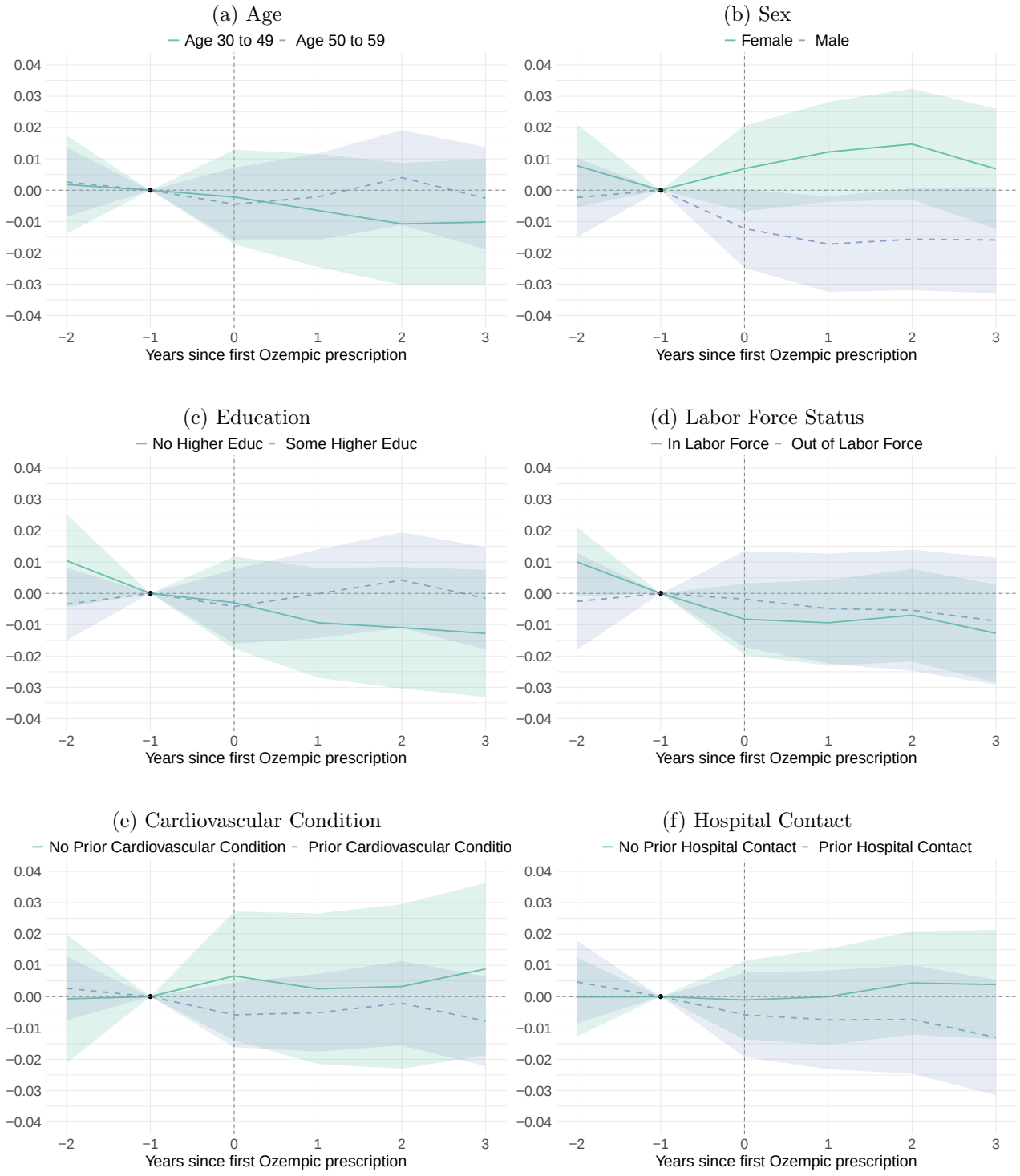
Notes: This figure plots robustness of our baseline matching specification for our four primary outcomes. Each panel displays four different specifications: i) baseline (as detailed in Section 4); ii) Callaway and Sant'Anna (2021) estimator and later-treated as comparison group (N=59,443 individuals); iii) matching on characteristics measured in the year prior to initiation instead of in 2016–2017 (N=14,008); and iv) restricting to the first four initiation cohorts (earlier treated in August 2018 – June 2019, N=10,830). Individuals are restricted to ages 30-59 across all robustness exercises. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. Baseline means over the pre-period are 3.789, 0.649, 0.677, and 12.387 in log (mean total income is DKK 286,430), for panels (a) to (d), respectively.

Appendix Figure B.7: Heterogeneity Analysis – Long-Term Sickness Leave, in Days



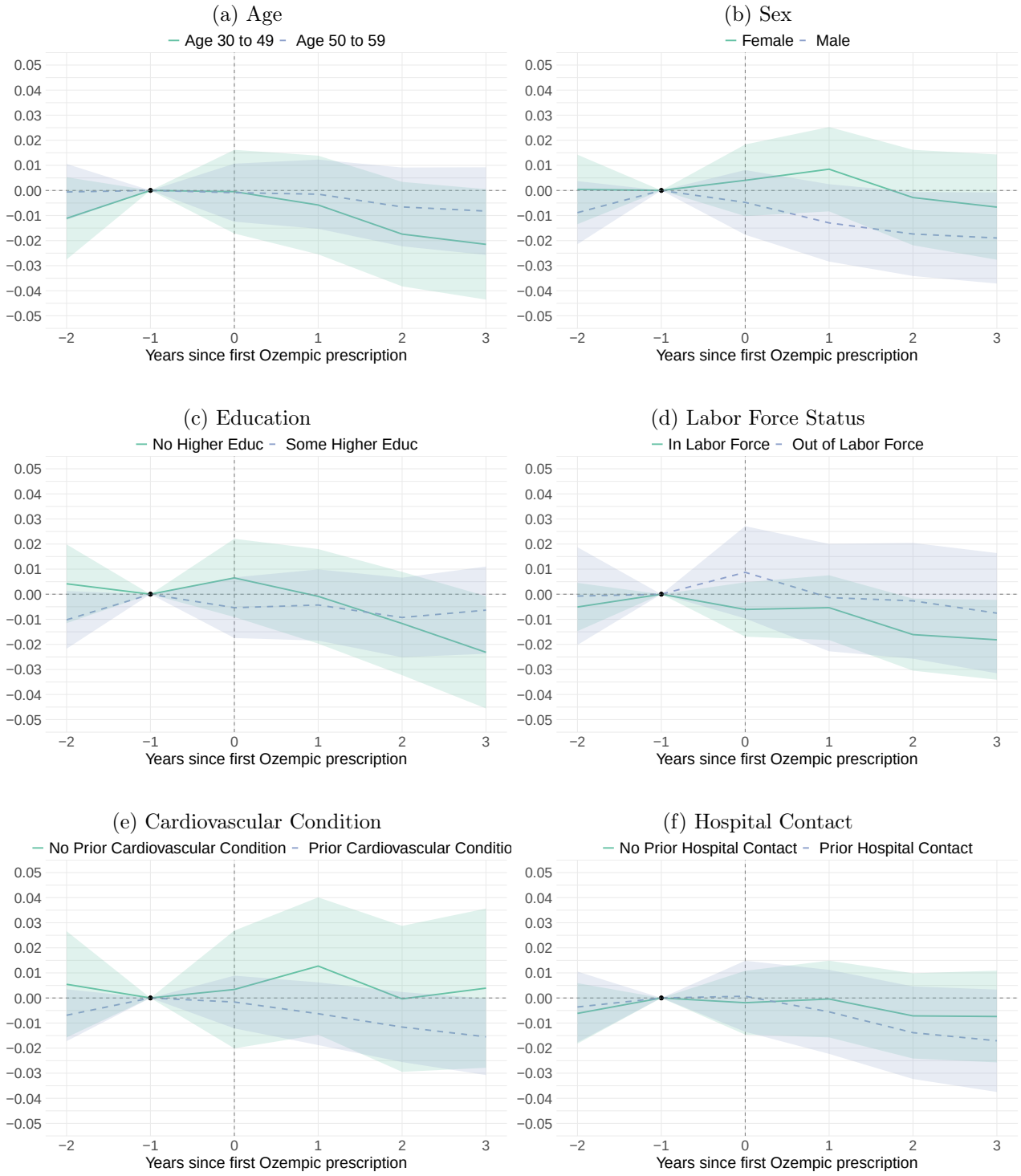
This figure plots the estimated event study coefficients from Equation (1) for long-term sickness leave for various heterogeneity margins. The aggregate difference-in-differences estimates are displayed in Figure 2. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Appendix Figure B.8: Heterogeneity Analysis – Employed



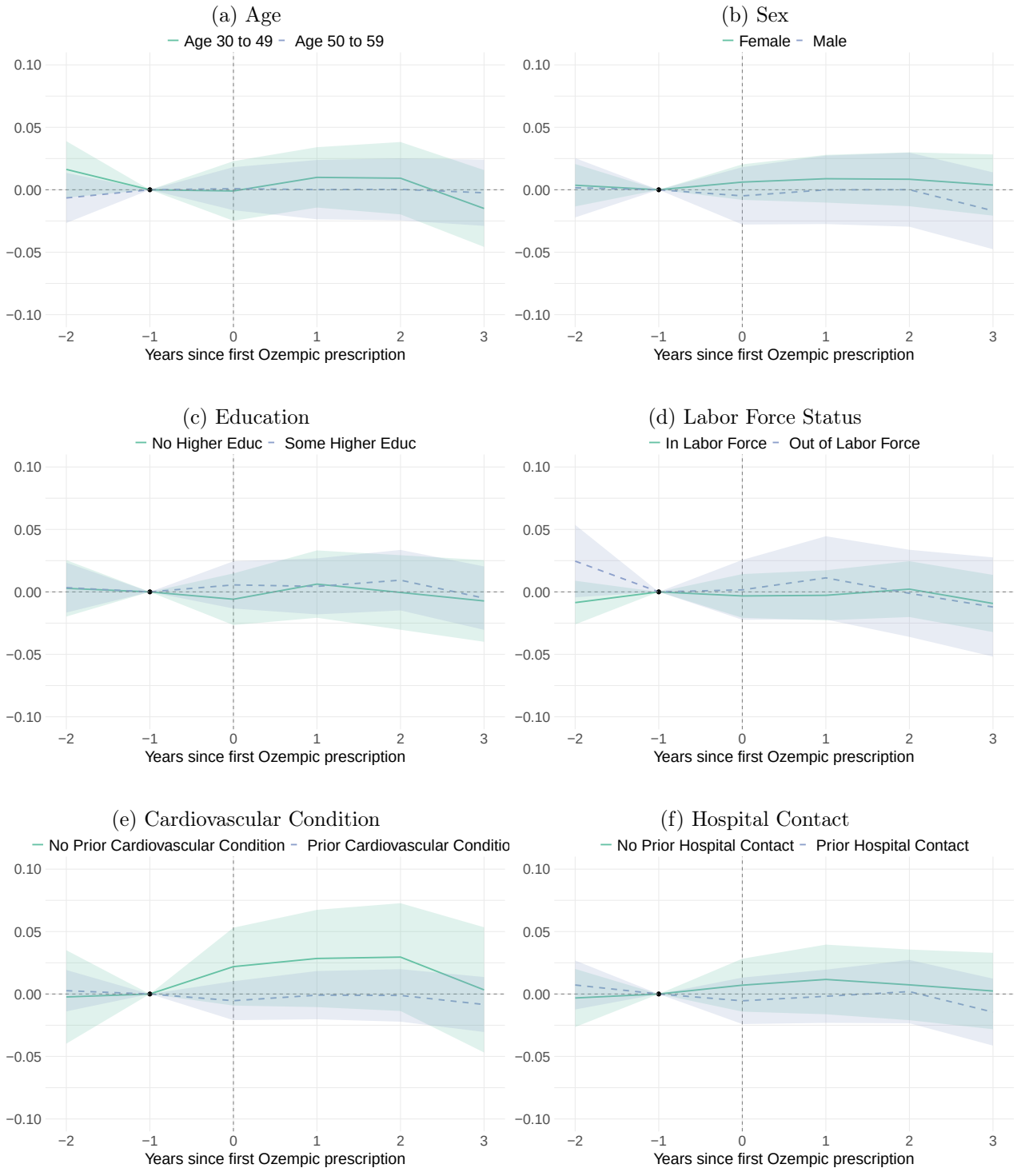
This figure plots the estimated event study coefficients from Equation (1) for being employed for various heterogeneity margins. The aggregate difference-in-differences estimates are displayed in Figure 2. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Appendix Figure B.9: Heterogeneity Analysis – In Labor Force



This figure plots the estimated event study coefficients from Equation (1) for being in the labor force for various heterogeneity margins. The aggregate difference-in-differences estimates are displayed in Figure 2. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Appendix Figure B.10: Heterogeneity Analysis – Log Total Income, Conditional on Positive Income



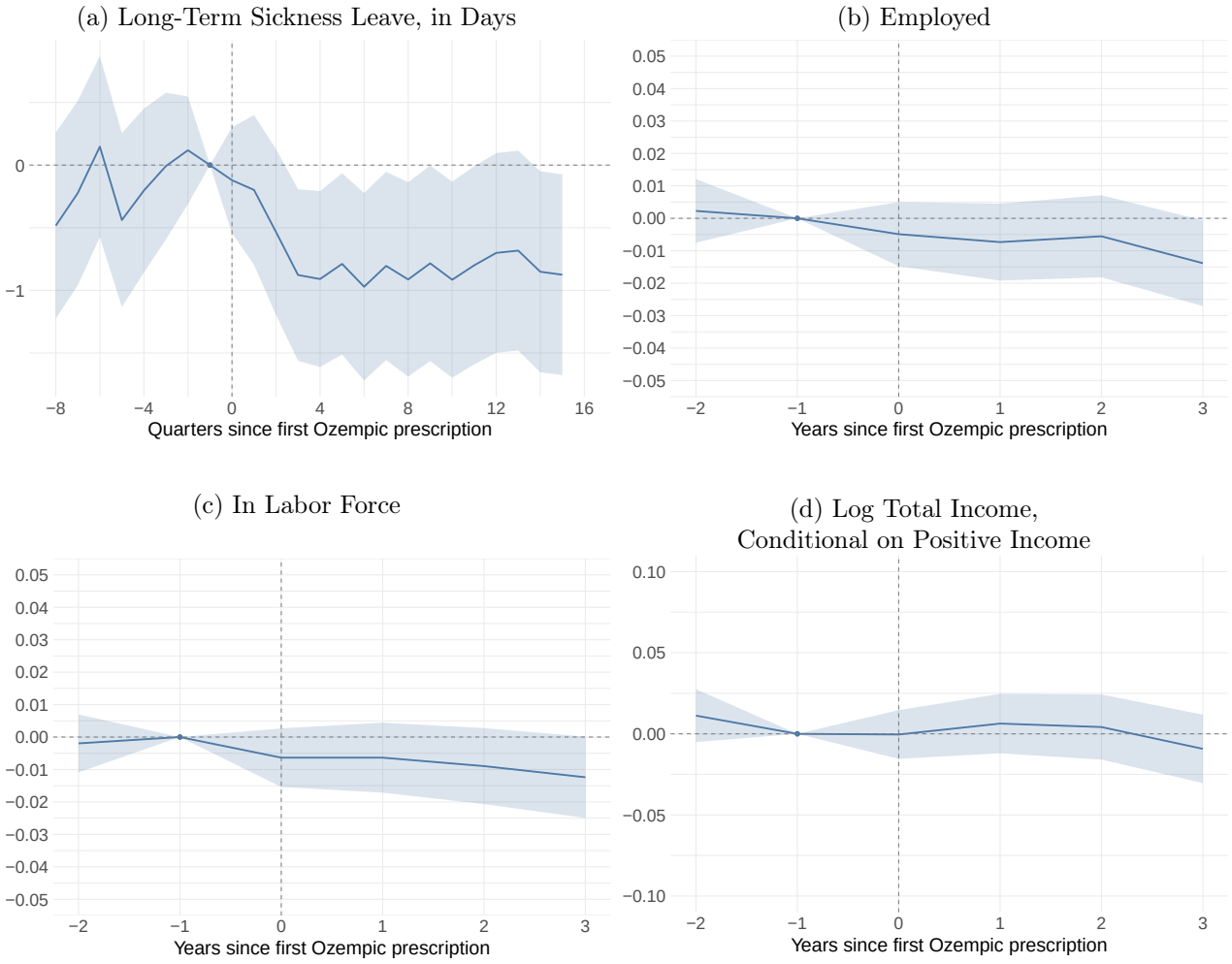
This figure plots the estimated event study coefficients from Equation (1) for log total income (conditional on positive income) for various heterogeneity margins. The aggregate difference-in-differences estimates are displayed in Figure 2. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

Appendix Figure B.11: Primary Outcomes for Women Not Employed at Baseline



Notes: This figure plots the estimated event study coefficients from Equation (1) for women who are not employed at baseline (in 2017) in dashed green and baseline estimates in black. This exercise is motivated by Diamond (2026) who finds large employment improvements for women (initiating GLP-1 for weight management) not employed at baseline. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed. N=1,072 early and 1,045 later initiators.

Appendix Figure B.12: Dropping Birth Cohorts Eligible for Early Retirement By 2023



Notes: This figure plots the estimated event study coefficients from Equation (1) for our primary outcomes, dropping individuals who are eligible for early retirement during our outcome horizon (*i.e.*, by the end of 2023). Specifically, we exclude birth cohorts 1958 and 1959 (see https://www.retsinformation.dk/eli/lta/2020/199 and https://www.retsinformation.dk/eli/lta/2022/527). The sample size now consists of 12,282 patients, of which exactly half are early initiators. The sample is restricted to individuals who survive throughout the four-year follow-up period. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

C Wegovy Initiation for Weight Management

This appendix estimates the impacts of Wegovy initiation on long-term sickness leave. Our main analysis focuses on Ozempic initiation, which is approved for diabetes management in Denmark rather than for weight management. Accordingly, 84 percent of individuals in our main analysis sample are diabetic at baseline. Wegovy provides a complementary treatment margin: in Denmark, Wegovy has been available since December 2022 and is the primary GLP-1 product approved specifically for weight management (Medicin.dk, 2026b). This treatment margin is also of policy importance, as GLP-1 use for weight management has expanded rapidly in recent years and is likely to account for an increasing share of future GLP-1 utilization.

Because Wegovy became available in Denmark only in December 2022 and our outcomes coverage ends in December 2023, the available follow-up period is too short to implement our baseline design, which matches initiators treated 16 quarters apart. We therefore use an alternative stacked difference-in-differences design following Diamond (2026)—who studies GLP-1 initiation for weight management in the U.S.—by comparing each initiation cohort to all later initiators observed before their own initiation. Additionally, the short follow-up period does not allow us to examine annual labor market outcomes. We therefore focus on treatment adherence and long-term sickness leave in this analysis.

Sample Definition. Our treatment group consists of individuals who initiate Wegovy between December 2022 and December 2023 and have no prior Ozempic use. This restriction isolates new Wegovy initiators from individuals switching from earlier Semaglutide treatment, so that initiation more closely captures the start of GLP-1 treatment for weight management and ensures that Wegovy initiation treatment effects are not contaminated by the causal impacts of prior Ozempic use.

We begin with 107,589 Wegovy initiators during this period. We further restrict the sample to men and women aged 30–59 whom we observe for at least four pre-initiation quarters. This results in a final analysis sample of 67,639 initiators. 70% of the sample are women and 8.2% had a diabetes diagnosis or diabetes drug use in the year prior to initiation.

Matching Procedure. Following Diamond (2026), we construct a stacked difference-in-differences design in which individuals initiating Wegovy in a given quarter are compared to individuals initiating Wegovy in later quarters. Later initiators serve as controls only in quarters before their own Wegovy initiation.

We adapt the matching procedure to the Danish register data. Diamond (2026) focuses on women and exactly matches individuals by partnership status and whether they are employed or on paid leave. Because our Wegovy analysis includes both men and women, we additionally exact match on sex. In the Danish registers, we define partnership as being married or in a registered partnership, and we define labor market attachment using whether an individual is in the labor force.

For each initiation cohort, we measure baseline characteristics in the quarter prior to (and year prior for labor market attachment) Wegovy initiation, and form exact-match cells based on sex, partnership status, and labor market attachment. Within each exact-match cell, we estimate propensity scores and re-weight later-initiating controls using propensity score odds weights. The covariates are chosen to capture the same broad dimensions as Diamond's survey-based controls, subject to availability in the Danish registers, and are described in Appendix Table C.1.

Stacked Difference-in-Differences Specification. Let G_i denote the quarter in which individual i initiates Wegovy. For each initiation cohort g, we construct a cohort-specific stack containing

individuals with $G_i = g$ and individuals who initiate Wegovy later. Later initiators serve as controls only in quarters before their own initiation. We estimate event study specifications of the form:

$$y_{igt} = \alpha_{ig} + \lambda_{gt} + \sum_{e \neq -1} \beta_e \mathbf{1}\{G_i = g\}\mathbf{1}\{t - g = e\} + \varepsilon_{igt}, \tag{2}$$

where α_{ig} are individual-by-stack fixed effects and λ_{gt} are initiation-cohort-by-calendar-quarter fixed effects. The omitted event time period is the quarter before Wegovy initiation. Observations are weighted using the matching and propensity score weights described above, and standard errors are clustered at the individual level.

The identifying comparison is between Wegovy initiators in cohort g and similar individuals who initiate Wegovy later but remain untreated in quarter t. Because initiation cohorts span only 2022Q4 through 2023Q4, later-initiator controls provide support for at most three quarters after initiation. We restrict the analysis to quarterly outcomes, as the short initiation window does not provide a valid comparison group for annual post-treatment outcomes. In particular, the only supported annual initiation year comparison is the December 2022 cohort, for which annual outcomes largely precede treatment.[1]

Results. We begin by estimating the first stage effect on continued Semaglutide use. Three quarters after Wegovy initiation, Semaglutide use declines to approximately 75% (Panel (a) of Appendix Figure C.1). This rate is lower than in our baseline Ozempic analysis, suggesting that patients who initiate Wegovy for weight management are less likely to remain on Semaglutide treatment over the longer run than patients who initiate Ozempic primarily for diabetes management.

Next, Panel (b) of Appendix Figure C.1 plots event study coefficients for long-term sickness leave days. By the third post-initiation quarter, Wegovy initiation is associated with a statistically insignificant reduction of approximately 0.63 days in long-term sickness leave. This estimate is similar in magnitude to our baseline estimates for Ozempic initiators in 2018–2019 (0.81 days per quarter over the two years in Table 2).

This Wegovy analysis complements our main Ozempic initiation design by focusing on the GLP-1 product approved for weight management in Denmark. The exercise differs from Diamond (2026)'s in three important ways. First, we use registry-based Wegovy initiation as a proxy for GLP-1 use for weight loss, whereas she observes self-reported GLP-1 use for weight loss. Second, our baseline covariates are administrative proxies for survey-based health and socioeconomic measures. Third, our comparison group is limited to later Wegovy initiators, not including interested never-users as controls. More broadly, Denmark and the United States are distinct institutional settings, and our results are not intended as a replication of Diamond (2026). A promising avenue for future research is to understand how institutional differences shape GLP-1 initiation and the mechanisms through which treatment affects labor market outcomes, including health, productivity, discrimination, and barriers to work.

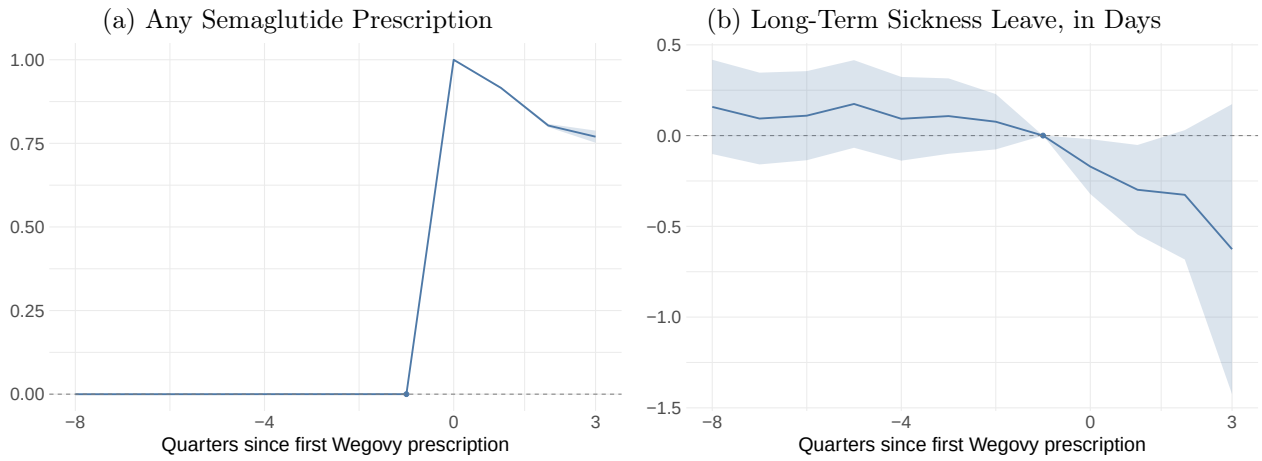
[1] Individuals initiating Wegovy in 2023 can serve as controls for the December 2022 cohort only for annual outcomes measured in 2022, since controls are retained only in calendar years preceding their own initiation. This corresponds to event year 0, when annual outcomes largely precede treatment, leaving no valid annual post-treatment comparison.

Appendix Table C.1: Matching Variables in Diamond (2026) and Our Wegovy Analysis

	Diamond (2026)	Our Analysis
Exact match:		
Sex	—	Sex
Partnership	Partnered	Married or registered partnership
Labor-market attachment	Employed or on paid leave	In labor force
Propensity score reweighting:		
Demographics	Race	First or second generation immigrant
Body weight	BMI	Obesity diagnosis
Diabetes	Diabetes status	Diabetic diagnosis or diabetes drug use
General health	Self-rated health	Number of GP visits, specialist visits, cardiovascular drug use
Mental health	Depression score	Antidepressant use
Work-limiting health	Whether health limits work	Disability insurance
Life satisfaction	Life satisfaction	—
Income	Household income	Individual total income
Cohort fixed effect	Cohort fixed effect	Cohort fixed effect

Notes: The table compares the variables used in our matching procedure with those used in Diamond (2026). Differences in variable definitions reflect the fact that our study uses Danish administrative registers instead of U.S. survey data. Diamond (2026)'s primary sample is restricted to women, although she also reports results separately by sex; in our pooled sample, we therefore include sex as an additional exact-match variable. The vast majority of individuals in the labor force who are not employed will receive some paid benefits in Denmark. All variables are taken at baseline in the quarter prior to initiation when available or in November of the year prior to initiation for labor market variables. Within each exact-match cell, later-initiating controls are reweighted using propensity score odds weights.

Appendix Figure C.1: Event Studies Around Wegovy Initiation



Notes: The figure reports event study estimates of Equation (2) on quarterly Semaglutide prescription and long-term sickness leave in days. Each panel plots coefficients from the stacked difference-in-differences specification described in Appendix C. The omitted event time period is the quarter before Wegovy initiation. Later Wegovy initiators serve as controls only in quarters before their own initiation. Estimates are weighted using the exact-match and propensity score odds weights described in Appendix Table C.1. 95% confidence intervals constructed from heteroskedastic standard errors clustered at the individual-level are displayed.

D Economic and Fiscal Benefits Calculation

This appendix provides the details of the economic benefit and fiscal spillover calculations of reduced sickness absences from Section 5.5. We use our primary outcome, calendar days of sickness leave in spells lasting more than 30 days, to value the reduction in absence at average daily earnings. We separately examine the effect on municipality-paid sickness benefits reported in Appendix Figure B.3.

We first value the reduction in long-term sickness absence. GLP-1 initiation reduces long-term sickness leave by 0.864 days per quarter, or 3.456 days annually, equivalent to 22.8% of the baseline mean. Valuing these days at calendar-day earnings based on average baseline annual labor income of DKK 304,616, and scaling by the employment share (0.65), yields DKK 4,437 per employed initiator per year, equivalent to 1.46% of annual labor earnings.[2]

We next examine public spending on sickness benefits paid directly to individuals after the first 30 days. The estimated annual reduction is DKK 545 per initiator, equivalent to DKK 839 when normalized by the employment share.[3] This economic benefit is already included in the overall reduction in long-term sickness leave days which includes the full spell conditional on lasting over 30 days.

Finally, we extend the valuation to short-term sickness absences. Assuming that the estimated 22.8% reduction in long-term sickness leave days also applies to days in spells lasting fewer than 30 days, the baseline of 2.784 short-term days per quarter implies 2.54 fewer days per employed worker annually. Valued at average calendar-day earnings, this corresponds to an additional annual benefit of DKK 2,119.[4]

The annual wage-valued benefit therefore rises from DKK 4,437, based only on long-term sickness absence, to DKK 6,557 once short-term absences are included, corresponding to 1.46% and 2.15% of annual labor earnings, respectively. These amounts are approximately USD 672 and USD 993. Applied to the average annual income of an American adult with diabetes, the same percentages imply annual values of approximately USD 826 to USD 1,221.[5]

To illustrate how these gains might be distributed, we assume that employers and municipalities together replace full wages throughout sickness spells and that sickness benefit measure captures all municipal savings associated with the avoided absences. This includes assuming that GLP-1 initiation does not affect municipal reimbursements to employers, which we do not observe. Thus, our calculation would represent a lower bound on the savings for municipalities. Under these assumptions, annual municipal savings of DKK 839 per employed initiator represent 12.8% of the combined long- and short-term benefit. The remaining DKK 5,718, or 87.2%, is allocated to employers.

[2] We assume that the treatment relative effect size on long-term sickness leave for all initiators is equivalent for all employed initiators.

[3] The benefit estimate is not statistically significant and therefore provides imprecise evidence of potential public savings.

[4] The short-term baseline corresponds to 11.136 calendar days annually and comes from supplemental absence data covering employees in the public sector and private firms with more than 250 employees (see Appendix Table B.2). The calculation is $4 \times 2.784 \times (0.864/3.789) \times (304{,}615.9/365)$. Applying this baseline to the broader sample assumes that it is representative of employed initiators.

[5] Authors' calculation using the 2023 Medical Expenditure Panel Survey for diabetic patients ages 30–59; average annual income is $56,720 in 2025 dollars.